

Untitled hardware project

Class ev_charger	Feasibility feasible	BoM Readiness Ready	Score 7.7
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Annual energy dispensed	219000	kWh/year	medium	DC output power multiplied by assumed equivalent full-power charging hours.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_intake_review_required

5/6 intake areas pass; 1 review, 0 blocked. Class ev_charger (high); requirements 1; numeric facts 1; requirement trace 1/1 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	182 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	review	0.5	1 quantified requirement(s), 1 numeric fact(s).	none	Proceed only as assumption-heavy review; ask for more quantified requirements before treating outputs as stable.
product_class_selection	pass	1	Selected ev_charger with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	ev_charger has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	1 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

clarification_recommended

4 clarification question(s): 0 required, 3 recommended, 1 optional. Architecture blockers 0. Next question: target_metrics:1.

#	Kind	Priority	Status	Question	Expected Answer	Example	Blocks Architecture
1	target_metrics	major	recommended	What are the top two or three quantified performance targets?	Metric name, value and unit for each target.	150 kW DC output, CCS2 connector, 95% charger efficiency.	no
2	operating_envelope	major	recommended	What operating envelope should the design satisfy?	A short list of dimensions, duty cycle, environment, runtime, capacity or output constraints.	Outdoor cabinet, 400 VAC input, depot environment, continuous 150 kW service.	no
3	interfaces_and_integration	major	recommended	What external systems, users or infrastructure must this design interface with?	List each external interface and any protocol, voltage, fluid, data or mechanical constraint.	Grid input, vehicle CCS2, OCPP backend, MID meter and emergency stop loop.	no
4	compliance_and_risk	minor	optional	Are there named standards, hazards or certification targets that must be treated as in-scope?	Named standard, jurisdiction, hazard or certification target.	IEC 61851, ISO 15118, MID metering, outdoor IP rating.	no

Architecture Admission Gate

architecture_generation_review_required

4/6 admission areas pass; 2 review, 0 blocked. Can use for review: yes; can proceed to BoM: no. Source scratch_universal_architecture; next action: Treat generated architecture as review-only until recommended clarification questions are answered..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	review	0.83	5/6 intake areas pass; class ev_charger (high); 1 quantified requirement(s).	none	Treat generated architecture as review-only until recommended clarification questions are answered.
clarification_clearance	review	1	4 clarification question(s): 0 required, 3 recommended, 0 architecture-blocking.	none	Keep the architecture labelled as review-only until recommended brief clarifications are answered.
class_grammar_clearance	pass	1	Product class ev_charger; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 10; submodules 30; components 120.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 6/6.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class ev_charger; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 259; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class ev_charger; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	507 provenance ref(s); 259 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	120 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufactur er/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifi er lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

1/6 freeze areas pass; 5 review, 0 blocked. Modules/submodules/components 10/30/120. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	review	0.88	Architecture admission architecture_generation_review_required; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean.
module_structure	review	0.6	Component allocation allocation_review_required; module engineering module_engineering_review_required; 0/10 modules pass; 7 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 6/6 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 1/1 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.5	Pre-BoM engineering engineering_review_ready; 3/6 areas pass; calculations needing review 1; assumptions needing review 23.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	120 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 52 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

5 closure row(s): 5 ready, 0 blocked. Queues S/E/V/R 1/1/2/1; next row freeze-close:1:admission_and_lineage. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	admission_and_lineage	architecture_review	minor	ready	Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean.	Architecture admission verdict is architecture_generation_admitted. Stage integrity verdict is stage_trace_accepted. Scratch lineage verdict is scratch_lineage_clean.	*.architecture-admission-gate.csv; *.stage-integrity-gate.csv; *.scratch-lineage-gate.csv
2	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
3	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
4	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
5	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.engineering-verification-plan.csv

Component Candidate Gate

component_candidates_review_required

6/7 candidate areas pass; 1 review, 0 blocked. Candidate lines 120; critical 10/10; worklist rows 120; duplicate groups 3; provenance violations 0; ready for sourcing no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	120 candidate BoM line(s), 117 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	120/120 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	120/120 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	10/10 class-critical part(s) are represented by candidate lines; 13 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	review	0.5	3 duplicate component identity group(s), 3 involving critical lines.	insulation_monitoring_device: 2 allocations need shared-vs-separate review. iso_15118_plc_mode_m: 2 allocations need shared-vs-separate review. mid_energy_meter: 2 allocations need shared-vs-separate review.	Resolve whether duplicate component IDs are shared physical items or separate install locations before pricing.
sourcing_worklist_alignment	pass	1	120/120 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorization_review_required

4/6 sourcing-authorization areas pass; 2 review, 0 blocked. Critical intake 13/13; full intake 120/120; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	review	0.5	architecture_generation_review_required; review yes, BoM no.	none	Keep sourcing as review-only until architecture admission is fully cleared.
component_candidate_authorization	review	0.86	component_candidates_review_required; 120 candidate line(s), 120 sourcing-worklist row(s), 0 provenance violation(s).	none	Resolve candidate review rows before admitting sourcing evidence.
critical_intake_scope	pass	1	13/13 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	120/120 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_review_required

Display mode candidate_only. 1/7 BoM-admission areas pass; 6 review, 0 blocked. Critical priced 0/13; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	review	0.67	sourcing_authorization_review_required ; authorized no; 4/6 areas pass.	none	Keep BoM admission review-only until sourcing authorization clears.
candidate_bom_scope	pass	1	120 candidate BoM line(s), 13 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/120 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/13 critical line(s) have admitted unit costs; 13 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.91	0 source-backed claim(s), 52 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 13. Verification accepted: 0/27; unaccepted: 27. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 13 critical BoM line(s) lack source-backed cost evidence.
- 7 verification activity/activities remain blocked.
- 0/27 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	13/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

3/6 areas pass; 3 review, 0 blocked. Modules/submodules/components: 10/30/120. Calculations within envelope 1/2; calculation blockers 0. Assumption review rows 23; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	10 modules, 30 submodules and 120 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	review	1	24/30 submodules ready; 10/10 critical parts allocated; 3 duplicate component groups.	none	Resolve duplicate component identity before sourcing: mid_energy_meter. Resolve duplicate component identity before sourcing: iso_15118_plc_mode m. Resolve duplicate component identity before sourcing: insulation_monitoring_device. Resolve whether these allocations are one shared physical item or separate install locations before sourcing, costing or deduplicating.
interface_verification	pass	1	6/6 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	1/1 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	review	0.5	1/2 calculations are within envelope; 1 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.23	7/30 assumptions are brief-supported or model-present; 23 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

0/7 costing areas pass; 0 review, 7 blocked. Critical priced lines 0/13; unpriced critical lines 13. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/120 BoM lines have admitted pricing; rejected source records 0.	power_distribution_ac_input_switchgear_ac_input_breaker_and_spd: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_dc_output_switchgear_dc_output_contactor_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_billing_metering_branch_mid_energy_meter: Resolve duplicate component identity before sourcing: decide whether this is one shared physical item or separate allocated lines. energy_conversion_transduction_rectifier_power_modules_power_module_stack: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. charging_connector_interface_ccs2_cable_connector_ccs2_liquid_cooled_cable_and_connector: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 8 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/13 critical lines have admitted unit costs.	power_distribution_ac_input_switchgear_ac_input_breaker_and_spd: AC input breaker and SPD power_distribution_dc_output_switchgear_dc_output_contactor_set: DC output contactor set power_distribution_billing_metering_branch_mid_energy_	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				meter: MID energy meter energy_conversion_transduction_rectifier_power_modules_power_module_stack: Power module stack charging_connector_interface_ccs2_cable_connector_ccs2_liquid_cooled_cable_and_connector: CCS2 liquid-cooled cable and connector plus 8 more	
source_backed_provenance	blocked	0.91	0 source-backed claims; 52 critical missing-source claims; 0 provenance violations.	power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_dc_output_switchgear_dc_output_contactor_set/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost	Keep supplier, manufacturer, MPN, unit cost and lead-time fields empty until source-backed evidence is admitted.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				evidence before BoM can pass. plus 47 more	
component_identity	blocked	0.98	3 duplicate component identity group(s); 3 include critical lines.	insulation_monitoring_device: Resolve whether these allocations are one shared physical item or separate install locations before sourcing, costing or deduplicating. iso_15118_plc_mode_m: Resolve whether these allocations are one shared physical item or separate install locations before sourcing, costing or deduplicating. mid_energy_meter: Resolve whether these allocations are one shared physical item or separate install locations before sourcing, costing or deduplicating.	Resolve duplicated critical component identities before admitting or deduplicating costs.
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.
source_reference_quality	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	13 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

42 missing-evidence row(s): 13 sourcing, 29 verification. Ready 39; blocked 0. Blocks BoM 13; blocks publish 42. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	ac_input_breaker_and_spd	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	dc_output_contactor_set	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_in take	mid_energy _meter	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
9	sourcing	blocker	ready_for_in take	power_mod ule_stack	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
10	sourcing	blocker	ready_for_in take	ccs2_liquid _cooled_ca ble_and_co nnector	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_intake	iso_15118_plc_modem	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_intake	ocpp_communications_gateway	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_intake	insulation_monitoring_device	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	sourcing	blocker	ready_for_in take	cooling_loo p_assembly	supplier_cat atalogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
15	sourcing	blocker	ready_for_in take	emergency_ stop_and_s afety_interlo ck_set	supplier_cat atalogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
16	verification	major	ready_for_in take	design_revi ew:power_d istribution	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
17	verification	major	ready_for_in take	calculation: power_distri bution	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict;	anonymous reviewer; unchecked model assertion;	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidenceRef; evidenceNote; reviewedAt	test-fixture:// review; source-evidence claims that belong in sourcing intake	
18	verification	major	ready_for_intake	design_review:energy_conversion_transduction	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	calculation:energy_conversion_transduction	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
20	verification	major	ready_for_intake	design_review:charging_connector_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
21	verification	major	ready_for_intake	calculation:charging_connector_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
22	verification	major	ready_for_intake	design_review:control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
23	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
24	verification	major	ready_for_intake	design_review:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	calculation:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
26	verification	major	ready_for_intake	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
27	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
28	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_intake	design_review:hmi ergonomics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
30	verification	major	ready_for_in take	interface_re view:power_ distribution: energy_con version_tran sduction:ac _input_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
31	verification	major	ready_for_in take	interface_re view:energy _conversion _transducti on:charging _connector _interface:d c_output_b us	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_in take	interface_re view:control _compute_c ommunicati on:charging _connector _interface:is o15118_plc	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
33	verification	major	ready_for_intake	interface_review:control_compute_communication:energy_conversion_transduction:control_buses	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
34	verification	major	ready_for_intake	interface_review:sensing_instrumentation:control_compute_communication:metering_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
35	verification	major	ready_for_intake	interface_review:environmental_interface:energy_conversion_transduction:coolant_loop	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
36	verification	major	ready_for_intake	compliance_review:IEC-61851	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
37	verification	major	ready_for_intake	compliance_review:IEC-62196-3	named_engineering_review	activityId; evidenceKind; reviewerName	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef ; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
38	verification	major	ready_for_intake	compliance_review:ISO-15118	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
39	verification	major	ready_for_intake	compliance_review:OCCPP-1.6J/2.0.1	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
40	verification	major	ready_for_intake	risk_review:1	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
41	verification	major	ready_for_intake	risk_review:2	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNote	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
42	verification	major	ready_for_in take	risk_review: 3	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/27. Source-backed BoM claims 0; critical missing BoM claims 52; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 40 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 65 blocked/source-required claim(s) and 107 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 7 review-required claim(s).
- power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- power_distribution_dc_output_switchgear_dc_output_contactor_set/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- power_distribution_dc_output_switchgear_dc_output_contactor_set/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- power_distribution_dc_output_switchgear_dc_output_contactor_set/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	10 modules, 30 submodules and 6 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0	1/225 claims pass; 67 blocked/source-required, 157 review-required.	architecture design has 0 blocked/source-required claim(s) and 40 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s). bom sourcing has 65 blocked/source-required claim(s) and 107 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 7 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/1 requirements have accepted assurance; 1 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/27 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 27 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.91	0 source-backed BoM claims, 52 critical missing-source claims, 0 provenance violations.	power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_ac_input_switchgear_ac_input_breaker	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				_and_spd/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_ac_input_switchgear_ac_input_breaker_and_spd/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_dc_output_switchgear_dc_output_contactor_set/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 47 more	

evidence_authenticity	review	0	0/0 evidence row(s) are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
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Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
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Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 52; trust blockers 65. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	13	56	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	9	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	29	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance-domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

225 claims: 1 brief-supplied, 0 reviewer-accepted, 0 source-backed, 3 calculated needing review, 154 generated needing review, 67 source-required, 0 blocked. Blocking: 67 BoM, 223 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	CCS2 liquid-cooled cable and connector / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CCS2 liquid-cooled cable and connector / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CCS2 liquid-cooled cable and connector / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CCS2 liquid-cooled cable and connector / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	OCPP communications gateway / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	OCPP communications gateway / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	OCPP communications gateway / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	OCPP communications gateway / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ISO 15118 PLC modem / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Power module stack / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Power module stack / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Power module stack / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Power module stack / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Cooling loop assembly / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Cooling loop assembly / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Cooling loop assembly / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Cooling loop	unit_cost_gbp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			assembly / unit_cost_g bp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	AC input breaker and SPD / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	AC input breaker and SPD / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	AC input breaker and SPD / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	AC input breaker and SPD / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MID energy meter / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	MID energy meter / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MID energy meter / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MID energy meter / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DC output contactor set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DC output contactor set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DC output contactor set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	DC output contactor set / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Emergency stop and safety interlock set /	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			manufactur er					cost evidence before BoM can pass.
bom_source _field	source_ required	assumption	Emergency stop and safety interlock set / mpn	mpn is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source _field	source_ required	assumption	Emergency stop and safety interlock set / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source _field	source_ required	assumption	Emergency stop and safety interlock set / unit_cost_g bp	unit_cost_g bp is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source _field	source_ required	assumption	MID energy meter / manufactur er	manufactur er is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source _field	source_ required	assumption	MID energy meter / mpn	mpn is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source _field	source_ required	assumption	MID energy meter / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MID energy meter / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Insulation monitoring device / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
component_candidate	source_required	generated_model	CCS2 liquid-	CCS2 liquid-	model:scratch_universa	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			cooled cable and connector	cooled cable and connector is an architecture-derived component candidate with quantity 1 candidate.	l_architecture.ev_charger			backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	ISO 15118 PLC modem	ISO 15118 PLC modem is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	OCPP communications gateway	OCPP communications gateway is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	ISO 15118 PLC modem	ISO 15118 PLC modem is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Power module stack	Power module stack is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								sourcing intake.
component_candidate	source_required	generated_model	Cooling loop assembly	Cooling loop assembly is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	AC input breaker and SPD	AC input breaker and SPD is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	MID energy meter	MID energy meter is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	DC output contactor set	DC output contactor set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Insulation monitoring device	Insulation monitoring device is an architecture-derived	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufactur

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				component candidate with quantity 1 candidate.				er, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Emergency stop and safety interlock set	Emergency stop and safety interlock set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	MID energy meter	MID energy meter is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Insulation monitoring device	Insulation monitoring device is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:ev_charger.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost metric as a claim.
component_candidate	generated_needs_review	generated_model	cable strain relief boot	cable strain relief boot is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	connector temperature sensor pair	connector temperature sensor pair is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	proximity pilot resistor set	proximity pilot resistor set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.ev_charger	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 225 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

1/225 claims pass by being brief-supplied, source-backed or reviewer-accepted. 67 blocked/source-required claims, 157 review-required claims. Claim pass ratio 0. Blocking: 67 BoM, 223 publishable.

- architecture design has 0 blocked/source-required claim(s) and 40 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 65 blocked/source-required claim(s) and 107 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 7 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	1	1	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	40	0	40	0	0	0	40	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	5	0	3	2	2	2	4	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	172	0	107	65	65	65	172	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	7	0	7	0	0	0	7	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

52 open evidence gaps: 16 blockers, 36 major, 0 minor. Closure paths: 13 sourcing intake, 29 verification intake, 9 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 13, publishable 52.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	13 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	7 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/27 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
calculation_review	major	needs review	EV charger daily full-power energy	600 kWh/day assumes 4 equivalent full-power hours.	engineering_review	publishable	Confirm expected sessions per day, dwell time, grid capacity, charger derating and tariff assumptions.
assumption_review	major	needs review	Annual energy dispensed	Annual energy dispensed uses $dc_power_kw * equivalent_full_power_hours_per_day * days_per_year$ with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	DC Charger Power Level	DC Charger Power Level currently evaluates as pass against 50 to 350 kW for common public DC fast-charger classes; >350 kW needs heavier cable, grid and cooling evidence.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Vehicle/Backend Protocol Closure	Vehicle/Backend Protocol Closure currently evaluates as	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
assumption_review	major	needs review	Power And Safety Chain	Power And Safety Chain currently evaluates as pass against AC input, DC output, cooling, insulation monitoring, emergency stop and protective earth present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
	major	needs review	IEC-61851	Conductive charging system requirements is relevant to this product class in UK/EU.	engineering_review	publishable	Charging modes, control pilot and safety-function evidence.
	major	needs review	IEC-62196-3	DC vehicle connector requirements is relevant to this product class in UK/EU.	engineering_review	publishable	CCS2 connector rating, cable assembly and temperature evidence.
	major	needs review	ISO-15118	Vehicle-to-grid communication interface is relevant to this product class in Global / EU.	engineering_review	publishable	PLC communication, Plug and Charge and session-state evidence where applicable.
	major	needs review	OCPP-1.6J/2.0.1	Charge point backend protocol is relevant to this product class in Operator requirement.	engineering_review	publishable	Backend session, metering, fault and firmware-management evidence.
sourcing_evidence	blocker	ready for intake	AC input breaker and SPD	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	DC output contactor set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	MID energy meter	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Power module stack	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	CCS2 liquid-cooled cable and connector	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	ISO 15118 PLC modem	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	OCPP communications gateway	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Insulation monitoring device	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Cooling loop assembly	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Emergency stop and safety interlock set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	AC Input And DC Output Distribution	Review AC Input And DC Output Distribution	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				architecture allocation			intake template.
verification_evidence	major	ready for intake	AC Input And DC Output Distribution	Verify AC Input And DC Output Distribution requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Conversion Stack	Review Power Conversion Stack architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Conversion Stack	Verify Power Conversion Stack requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Vehicle Charging Interface	Review Vehicle Charging Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Vehicle Charging Interface	Verify Vehicle Charging Interface requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication	Review Control Compute Communication architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Metering And Electrical Sensing	Review Metering And Electrical Sensing architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thermal And Environmental Interface	Review Thermal And Environmental Interface	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				architecture allocation			intake template.
verification_evidence	major	ready for intake	Thermal And Environmental Interface	Verify Thermal And Environmental Interface requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Cabinet Structure And Containment	Review Cabinet Structure And Containment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Safety Protection	Review Safety Protection architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Maintenance And Commissioning	Review Maintenance And Commissioning architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	User Payment And HMI	Review User Payment And HMI architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	AC Input And DC Output Distribution -> Power Conversion Stack	Verify ac_input_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Conversion Stack -> Vehicle Charging Interface	Verify dc_output_buses interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication -> Vehicle	Verify iso15118_plc interface contract	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
			Charging Interface				intake template.
verification_evidence	major	ready for intake	Control Compute Communication -> Power Conversion Stack	Verify control_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Metering And Electrical Sensing -> Control Compute Communication	Verify metering_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thermal And Environmental Interface -> Power Conversion Stack	Verify coolant_loop interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for IEC-61851	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for IEC-62196-3	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for ISO-15118	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for OCPP-1.6J/2.0.1	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for DC output fault or insulation failure	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Connector overheating at high current	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Backend or payment outage stops sessions	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 13, engineering 9, verification 29, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	13	13/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:ac_input_breaker_and_spd; sourcing:ccs2_liquid_cooled_cable_and_connector; sourcing:cooling_loop_assembly; sourcing:dc_output_contactor_set; sourcing:emergency_stop_and_safety_interlock_set; plus 5 more
3	Complete Engineering Review	9	0/9/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:IEC-61851; assumption:compliance:IEC-62196-3; assumption:compliance:ISO-15118; assumption:compliance:OCPP-1.6J/2.0.1; assumption:metric:headline_output; assumption:sanity:evcharger_power_level; assumption:sanity:evcharger_power_safety_chain; assumption:sanity:evcharger_protocol_closure; plus 1 more
4	Accept Reviewer Evidence	29	2/27/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation:charging_connector_interface; verification:calculation:energy_conversion_transduction;

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							verification:calculation:environmental_interface; verification:calculation:power_distribution; verification:compliance_review:IEC-61851; verification:compliance_review:IEC-62196-3; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

5 meeting benchmark, 2 below benchmark, average comparable ratio 1.04.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	10 modules	9 modules	1.11	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	30 submodules	27 submodules	1.11	meets	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	120 components	108 components	1.11	meets	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	48 links	34 links	1.41	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	34 activities	20 activities	1.7	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a 150 kW DC fast EV charger with CCS2 liquid-cooled cable, OCPP backend, ISO 15118 PLC communication, MID metering, insulation monitoring, emergency stop, and outdoor cabinet.

Requirement	Value	Unit
DC output power	150	kW

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
DC output power	covered	150 kW	AC Input And DC Output Distribution / AC input switchgear; Power Conversion Stack / Rectifier power modules; Vehicle Charging Interface / CCS2 cable connector; Thermal And Environmental Interface / Liquid cooling loop	metric:headline_out put; sanity:evcharger_po wer_level; sanity:evcharger_po wer_safety_chain; sanity:evcharger_pr otocol_closure	DC power is traced through input/output distribution, conversion stack, connector interface and thermal support. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

1 requirement row(s): 0 accepted with evidence, 1 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
DC output power	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_acti vity	accepted_verific ation	power_distributi on; energy_conversi on_transduction; charging_conne ctor_interface; environmental_i nterface	design_review:p ower_distributio n; calculation:powe r_distribution; design_review:e nergy_conversio n_transduction; calculation:ener gy_conversion_t ransduction; design_review:c harging_connect or_interface; calculation:char ging_connector_ interface; design_review:e nvironmental_int erface; calculation:envir onmental_interfa ce	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Engineering Assurance Matrix

1 requirements: 0 accepted, 0 ready, 1 needing review, 0 blocked, 0 unlinked. 1 have calculation links, 1 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
DC output power	needs review	covered	AC Input And DC Output Distribution; Power Conversion Stack; Vehicle Charging Interface; Thermal And Environmental Interface	ev_charger_800v_current_a; ev_charger_daily_full_power_energy_kwh	assumption_resolution: component: charging_connector_interface_ccs2_cable_connector_ccs2_liquid_cooled_cable_and_connector; assumption_resolution: component: charging_connector_interface_vehicle_communication_frontend_iso_15118_plc_mode_m; assumption_resolution: component: control_computation_communication_protocol_stack_iso_15118_plc_mode_m; assumption_resolution: component: energy_conversion_transduction_rectifier_power_modules_power_module_stack; assumption_resolution: component: environmental_interface_liquid_cooling_loop_cooling_loop_assembly; plus 30 more	pending; pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 1; numeric_fact_count : 1; requirement_trace_covered: 1; requirement_trace_partial: 0; requirement_trace_uncovered: 0	dc_power_kw: 150 kW; dc_power_kw: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected ev_charger with high confidence from brief keywords.	selected_class: ev_charger; confidence: high; keyword_scores: ev_charger:11	ev_charger: 11	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 10-module ev_charger architecture from the scratch universal architecture grammar.	module_count: 10; architecture_source : scratch_universal_architecture	AC Input And DC Output Distribution; Power Conversion Stack; Vehicle Charging Interface; Control Compute Communication; Metering And Electrical Sensing; Thermal And Environmental Interface; plus 4 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 30; average_submodules_per_module: 3	power_distribution.ac_input_switchgear; power_distribution.dc_output_switchgear; power_distribution.billing_metering_branch; energy_conversion_transduction.rectifier_power_modules; energy_conversion_transduction.output_filtering; energy_conversion_transduction.converter_supervision; plus 24 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 6; present_links: 6; missing_links: 0; graph_nodes: 40; graph_edges: 78; shared_interface_edges: 42; missing_required_graph_edges: 0	power_distribution -> energy_conversion_transduction via ac_input_bus: present; energy_conversion_transduction -> charging_connector_interface via dc_output_bus:	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
				present; control_compute_c ommunication -> charging_connector _interface via iso15118_plc: present; control_compute_c ommunication -> energy_conversion_ transduction via control_bus: present; sensing_instrument ation -> control_compute_c ommunication via metering_bus: present; environmental_inter face -> energy_conversion_ transduction via coolant_loop: present	
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 120; critical_candidate_li nes: 13; admitted_price_line s: 0	ac_input_breaker_a nd_spd: AC input breaker and SPD; lockable_ac_isolato r: lockable AC isolator; surge_arrestor_cart ridge: surge arrestor cartridge; input_terminal_shro ud: input terminal shroud; dc_output_contacto r_set: DC output contactor set; precharge_resistor_ module: precharge resistor module; plus 24 more	Showing first 30 of 120 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_cou nt: 0; sanity_pass: 3; sanity_warn: 0; sanity_fail: 0	pass DC Charger Power Level: 150 kW; pass Vehicle/Backend Protocol Closure: present; pass Power And Safety Chain: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign- off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 120; admitted_lines: 0; unpriced_lines: 120; unpriced_critical_lin es: 13; rejected_source_re cords: 0	none	BoM cost remains intentionally blocked until 13 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

10 modules, 30 sub-modules, 120 allocated component words.

From	To	Interface	Status	Engineering Reason
power_distribution	energy_conversion_transduction	ac_input_bus	OK	Grid input switchgear must feed the power-conversion stack over a protected AC bus.
energy_conversion_transduction	charging_connector_interface	dc_output_bus	OK	Power modules must feed the CCS2 connector over a protected DC output bus.
control_compute_communication	charging_connector_interface	iso15118_plc	OK	Charge controller must communicate with the vehicle for ISO 15118 / PLC handshakes.
control_compute_communication	energy_conversion_transduction	control_bus	OK	Charge controller must command power module output states.
sensing_instrumentation	control_compute_communication	metering_bus	OK	Energy and safety measurements must feed session control and billing.
environmental_interface	energy_conversion_transduction	coolant_loop	OK	Power conversion heat must couple into the charger cooling loop.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
10	30	42	6	0

Interface Contract Matrix

6/6 required contracts present. 15 shared interface families, 4 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
ac_input_bus	AC Input And DC Output Distribution	AC input switchgear	Power Conversion Stack	Rectifier power modules	present	Grid input switchgear must feed the power-conversion stack over a protected AC bus.	Both endpoints expose the required interface through at least one submodule carrier.
dc_output_buses	Power Conversion Stack	Rectifier power modules; Output filtering	Vehicle Charging Interface	CCS2 cable connector	present	Power modules must feed the CCS2 connector over a protected DC output bus.	Both endpoints expose the required interface through at least one submodule carrier.
iso15118_plc	Control Compute Communication	Vehicle protocol stack	Vehicle Charging Interface	Vehicle communication frontend	present	Charge controller must communicate with the vehicle for ISO 15118 / PLC handshakes.	Both endpoints expose the required interface through at least one submodule carrier.
control_bus	Control Compute Communication	Charge controller; Vehicle protocol stack	Power Conversion Stack	Rectifier power modules; Converter supervision	present	Charge controller must command power module output states.	Both endpoints expose the required interface through at least one submodule carrier.
metering_bus	Metering And Electrical Sensing	Energy metering	Control Compute Communication	Charge controller; Vehicle protocol stack	present	Energy and safety measurements must feed session control and billing.	Both endpoints expose the required interface through at least one submodule carrier.
coolant_loop	Thermal And Environmental Interface	Liquid cooling loop; Thermal air path	Power Conversion Stack	Rectifier power modules; Converter supervision	present	Power conversion heat must couple into the charger cooling loop.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: airflow_path (Thermal And Environmental Interface); condensate_drain (Thermal And Environmental Interface); emergency_stop (Safety Protection); mechanical_mounts (Cabinet Structure And Containment).

Interface Verification Gate

interface_review_ready

6 required interface row(s): 0 accepted, 6 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
ac_input_bus	AC Input And DC Output Distribution	Power Conversion Stack	both_endpoints_carried	interface_review:power_distribution:energy_conversion_transduction:ac_input_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
dc_output_buses	Power Conversion Stack	Vehicle Charging Interface	both_endpoints_carried	interface_review:energy_conversion_transduction:charging_connector_interface:dc_output_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
iso15118_plc	Control Compute Communication	Vehicle Charging Interface	both_endpoints_carried	interface_review:control_compute_communication:charging_connector_interface:iso15118_plc	pending	review_ready	Collect named interface-review evidence through verification intake.
control_bus	Control Compute Communication	Power Conversion Stack	both_endpoints_carried	interface_review:control_compute_communication:energy_conversion_transduction:control_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
metering_bus	Metering And Electrical Sensing	Control Compute Communication	both_endpoints_carried	interface_review:sensing_instrumentation:control_compute_communication:metering_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
coolant_loop	Thermal And Environmental Interface	Power Conversion Stack	both_endpoints_carried	interface_review:environmental_interface:energy_conversion_transduction:coolant_loop	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_review_required

24/30 submodules ready; 6 review, 0 blocked. Component words 120; critical parts 10/10; duplicate groups 3. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
AC Input And DC Output Distribution	AC input switchgear	ready	4	ac_input_bus; protective_earth; service_access	ac_input_breaker_and_spd	none	Submodule has component candidates and no allocation blocker.
AC Input And DC Output Distribution	DC output switchgear	ready	4	dc_output_buses; control_bus; protective_earth	dc_output_contactor_set	none	Submodule has component candidates and no allocation blocker.
AC Input And DC Output Distribution	Billing metering branch	review	4	metering_bus; dc_output_buses; service_access	mid_energy_meter	mid_energy_meter	Resolve duplicate component identity before sourcing: mid_energy_meter.
Power Conversion Stack	Rectifier power modules	ready	4	ac_input_bus; dc_output_buses; coolant_loop; control_bus	power_module_stack	none	Submodule has component candidates and no allocation blocker.
Power Conversion Stack	Output filtering	ready	4	dc_output_buses; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Power Conversion Stack	Converter supervision	ready	4	control_bus; sensor_bus; coolant_loop	none	none	Submodule has component candidates and no allocation blocker.
Vehicle Charging Interface	CCS2 cable connector	ready	4	dc_output_buses; ccs2_vehicle_interface; control_pilot; protective_earth	ccs2_liquid_cooled_cable_and_connector	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Vehicle Charging Interface	Connector cooling coupling	ready	4	coolant_loop; ccs2_vehicle_interface; service_access	none	none	Submodule has component candidates and no allocation blocker.
Vehicle Charging Interface	Vehicle communication frontend	review	4	iso15118_plc; control_pilot; control_bus	iso_15118_plc_modem	iso_15118_plc_modem	Resolve duplicate component identity before sourcing: iso_15118_plc_modem.
Control Compute Communication	Charge controller	ready	4	control_bus; metering_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Backend gateway	ready	4	ocpp_network ; service_port	ocpp_communications_gateway	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Vehicle protocol stack	review	4	iso15118_plc; control_bus; metering_bus	iso_15118_plc_modem	iso_15118_plc_modem	Resolve duplicate component identity before sourcing: iso_15118_plc_modem.
Metering And Electrical Sensing	Energy metering	review	4	metering_bus; dc_output_buses	mid_energy_meter	mid_energy_meter	Resolve duplicate component identity before sourcing: mid_energy_meter.
Metering And Electrical Sensing	Insulation fault detection	review	4	insulation_monitoring; dc_output_buses; alarm_bus	insulation_monitoring_device	insulation_monitoring_device	Resolve duplicate component identity before sourcing: insulation_monitoring_device.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Metering And Electrical Sensing	Connector sensing	ready	4	sensor_bus; ccs2_vehicle_interface; control_pilot	none	none	Submodule has component candidates and no allocation blocker.
Thermal And Environmental Interface	Liquid cooling loop	ready	4	coolant_loop; sensor_bus; service_accesses	cooling_loop_assembly	none	Submodule has component candidates and no allocation blocker.
Thermal And Environmental Interface	Thermal air path	ready	4	airflow_path; coolant_loop; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Thermal And Environmental Interface	Weatherproofing	ready	4	condensate_drain; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Cabinet Structure And Containment	Outdoor cabinet	ready	4	mechanical_mounts; service_accesses; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Cabinet Structure And Containment	Cable holster and boom	ready	4	ccs2_vehicle_interface; mechanical_mounts	none	none	Submodule has component candidates and no allocation blocker.
Cabinet Structure And Containment	Internal compartmentation	ready	4	service_accesses; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Safety Protection	Emergency stop chain	ready	4	emergency_stop; alarm_bus; control_bus	emergency_stop_and_safety_interlock_set	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Safety Protection	DC fault protection	review	4	dc_output_buss; insulation_monitoring; alarm_bus	insulation_monitoring_device	insulation_monitoring_device	Resolve duplicate component identity before sourcing: insulation_monitoring_device.
Safety Protection	User access protection	ready	4	protective_earth; ccs2_vehicle_interface	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Commissioning	Commissioning access	ready	4	service_accesses; service_port; metering_bus	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Commissioning	Replaceable modules	ready	4	service_accesses; control_bus; coolant_loop	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Commissioning	Remote diagnostics	ready	4	ocpp_network ; service_port; control_bus	none	none	Submodule has component candidates and no allocation blocker.
User Payment And HMI	User display panel	ready	4	control_bus; ocpp_network	none	none	Submodule has component candidates and no allocation blocker.
User Payment And HMI	Payment and identity	ready	4	ocpp_network ; control_bus	none	none	Submodule has component candidates and no allocation blocker.
User Payment And HMI	User cable guidance	ready	4	ccs2_vehicle_interface; service_accesses	none	none	Submodule has component candidates and no

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
							allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/30 submodules pass; 30 review, 0 blocked. Component words 120; linked requirements 4; carrier contracts 18/18; local-only interfaces 64. Review questions accepted 0/78; verification accepted 0/80; critical unpriced lines 19.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
AC Input And DC Output Distribution	AC input switchgear	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): protective_earth, service_access. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await acceptance. 1 evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
AC Input And DC Output Distribution	DC output switchgear	review	4	3	0	0/0	0/2	0/2	1	0.6667	Review 3 local-only interface declaration(s): dc_output_bus, control_bus, protective_earth. 2 engineering review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
AC Input And DC Output Distribution	Billing metering branch	review	4	3	0	0/0	0/2	0/5	2	0.6667	Resolve duplicate component identity before sourcing: mid_energy_meter. Review 3 local-only interfaces declaration(s): metering_bus, dc_output_bus, service_access. 2 engineering review question(s) await acceptance. 3 verification activity row(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await accepted evidence. 2 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Power Conversion Stack	Rectifier power modules	review	4	4	1	4/4	0/6	0/7	1	0.6667	6 engineering review question(s) await acceptance. 6 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Power Conversion Stack	Output filtering	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): protective_earth. 3 engineering review

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Power Conversion Stack	Converter supervision	review	4	3	0	2/2	0/4	0/3	0	0.6667	Review 1 local-only interface declaration(s): sensor_bus. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence.
Vehicle Charging Interface	CCS2 cable connector	review	4	4	1	1/1	0/3	0/4	1	0.6667	Review 3 local-only interface declaration(s): ccs2_vehicle_interface, control_pilot, protective_earth. 3 engineering review question(s) await acceptance.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											nce. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Vehicle Charging Interface	Connector cooling coupling	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): coolant_loop, ccs2_vehicle_interface, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Vehicle Charging Interface	Vehicle communication frontend	review	4	3	0	1/1	0/3	0/7	2	0.6667	Resolve duplicate component identity before

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											sourcing: iso_15118_plc_modem. Review 2 local-only interface declaration(s): control_pilot, control_bus. 3 engineering review question(s) await acceptance. 5 verification activity row(s) await accepted evidence. 2 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Control Computer Communication	Charge controller	review	4	3	0	2/2	0/4	0/3	0	0.6667	Review 1 local-only interface declaration(s): alarm_bus. 4 engineering review question(s) await acceptance. 3

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Control Compute Communication	Backend gateway	review	4	2	0	0/0	0/2	0/2	1	0.6667	verification activity row(s) await accepted evidence.
											Review 2 local-only interface declaration(s): ocpp_network, service_port. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Control Compute Communication	Vehicle protocol stack	review	4	3	0	3/3	0/5	0/7	2	0.6667	Resolve duplicate component identity before sourcing: iso_15118_plc_modem.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											5 engineering review question(s) await acceptance. 5 verification activity row(s) await accepted evidence. 2 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Metering And Electrical Sensing	Energy metering	review	4	2	0	1/1	0/3	0/5	2	0.6667	Resolve duplicate component identity before sourcing: mid_energy_meter. Review 1 local-only interface declaration(s): dc_output_bus. 3 engineering review question(s) await acceptance. 3 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence. 2 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Metering And Electrical Sensing	Insulation fault detection	review	4	3	0	0/0	0/2	0/4	2	0.6667	Resolve duplicate component identity before sourcing: insulation_monitoring_device. Review 3 local-only interfaces declaration(s): insulation_monitoring, dc_output_bus, alarm_bus. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											e. 2 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Metering And Electrical Sensing	Connect or sensing	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): sensor_bus, ccs2_vehicle_interface, control_pilot. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Thermal And Environmental Interface	Liquid cooling loop	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): sensor_bus, service_access. 3 engineering review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Thermal And Environmental Interface	Thermal air path	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): airflow_path, protective_earth. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Thermal And Environmental Interface	Weather proofing	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): conden

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											sate_drain, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Cabinet Structure And Containment	Outdoor cabinet	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): mechanical_mounts, service_access, protective_earth. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Cabinet Structure And Containment	Cable holster and boom	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): ccs2_ve

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											hicle_interface, mechanical_mounts. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Cabinet Structure And Containment	Internal compartmentation	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, protective_earth. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Safety Protection	Emergency stop chain	review	4	3	0	0/0	0/2	0/2	1	0.6667	Review 3 local-only interface declaration(s): emergency_stop, alarm_b

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											us, control_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Safety Protection	DC fault protection	review	4	3	0	0/0	0/2	0/4	2	0.6667	Resolve duplicate component identity before sourcing: insulation_monitoring_device. Review 3 local-only interface declaration(s): dc_output_bus, insulation_monitoring, alarm_bus. 2 engineering review

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 2 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Safety Protection	User access protection	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): protective_earth, ccs2_vehicle_interface. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Maintenance And	Commissioning access	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Commissioning											e declaration(s): service_access, service_port, metering_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Maintenance And Commissioning	Replaceable modules	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, control_bus, coolant_loop. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Maintenance And Commissioning	Remote diagnostics	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): ocpp_network, service_port, control_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
User Payment And HMI	User display panel	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): control_bus, ocpp_network. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
User Payment And HMI	Payment and identity	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): ocpp_network, control_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
User Payment And HMI	User cable guidance	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): ccs2_vehicle_interface, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

module_engineering_review_required

0/10 modules pass; 10 review, 0 blocked. Submodules/components: 30/120. Interface carriers 12/12; review questions accepted 0/91; critical unpriced lines 13 across 7 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
AC Input And DC Output Distributio n	review	3	12	1	1/1	0/14	3	0.86	3 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 10 assumptio n row(s) need review or source evidence.
Power Conversio n Stack	review	3	12	1	4/4	0/14	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 8 assumptio n row(s) need review or source evidence.
Vehicle Charging Interface	review	3	12	1	2/2	0/14	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 8 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Control Compute Communic ation	review	3	12	0	3/3	0/10	2	0.71	2 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Metering And Electrical Sensing	review	3	12	0	1/1	0/9	2	0.71	2 critical BoM line(s) need source- backed cost evidence. 9 engineerin g review question(s) await acceptanc e. 7 assumptio n row(s) need review or source evidence.
Thermal And Environme ntal Interface	review	3	12	1	1/1	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 5 assumptio n row(s) need

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									review or source evidence.
Cabinet Structure And Containme nt	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e.
Safety Protection	review	3	12	0	0/0	0/7	2	0.71	2 critical BoM line(s) need source- backed cost evidence. 7 engineerin g review question(s) await acceptanc e. 4 assumptio n row(s) need review or source evidence.
Maintenan ce And Commissi oning	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 2 assumptio n row(s) need review or source evidence.
User Payment And HMI	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 1 assumptio n row(s) need review or source evidence.

3 ready, 7 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
AC Input And DC Output Distribution	sourcing	3	12	1/1	dc_power_kw	ac_input_breaker_and_spd; dc_output_contactor_set; mid_energy_meter	3 critical line(s) need source-backed evidence before BoM pass.
Power Conversion Stack	sourcing	3	12	4/4	dc_power_kw	power_module_stack	1 critical line(s) need source-backed evidence before BoM pass.
Vehicle Charging Interface	sourcing	3	12	2/2	dc_power_kw	ccs2_liquid_cooled_cable_and_connector; iso_15118_plc_modem	2 critical line(s) need source-backed evidence before BoM pass.
Control Compute Communication	sourcing	3	12	3/3	none	ocpp_communications_gateway; iso_15118_plc_modem	2 critical line(s) need source-backed evidence before BoM pass.
Metering And Electrical Sensing	sourcing	3	12	1/1	none	mid_energy_meter; insulation_monitoring_device	2 critical line(s) need source-backed evidence before BoM pass.
Thermal And Environmental Interface	sourcing	3	12	1/1	dc_power_kw	cooling_loop_assembly	1 critical line(s) need source-backed evidence before BoM pass.
Cabinet Structure And Containment	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Safety Protection	sourcing	3	12	0/0	none	emergency_stop_and_safety_interlock_set; insulation_monitoring_device	2 critical line(s) need source-backed evidence before BoM pass.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Maintenance And Commissioning	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
User Payment And HMI	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Engineering Review Pack

77 reviewer questions: 0 accepted, 47 ready, 30 needing review, 0 blocked. Coverage: 10 modules, 30 submodules, 6 interfaces, 2 calculations, 29 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumption_resolution	major	needs review	IEC-61851	Can the IEC-61851 assumption be accepted for architecture review, or must it be replaced with evidence?	Charging modes, control pilot and safety-function evidence.	none	none	none	none
assumption_resolution	major	needs review	IEC-62196-3	Can the IEC-62196-3 assumption be accepted for architecture review, or must it be replaced with evidence?	CCS2 connector rating, cable assembly and temperature evidence.	none	none	none	none
assumption_resolution	major	needs review	ISO-15118	Can the ISO-15118 assumption be accepted for architecture review, or must it be replaced with evidence?	PLC communication, Plug and Charge and session-state evidence where applicable.	none	none	none	none
assumption_resolution	major	needs review	OCPP-1.6J/2.0.1	Can the OCPP-1.6J/2.0.1 assumption be accepted for architecture review, or must it be replaced with evidence?	Backend session, metering, fault and firmware-management evidence.	none	none	none	none
assumption_resolution	major	needs review	CCS2 liquid-cooled cable and connector	Can the CCS2 liquid-cooled cable and connector	Admit supplier, manufacturer, MPN where applicable,	charging_connector_interface	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				assumptio n be accepted for architectu re review, or must it be replaced with evidence?	unit cost, lead time and evidence reference through sourcing intake.				
assumptio n_resoluti on	major	needs review	ISO 15118 PLC modem	Can the ISO 15118 PLC modem assumption be accepted for architectu re review, or must it be replaced with evidence?	Admit supplier, manufactu rer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	charging_ connector _interface; control_co mpute_co mmunicati on	none	none	none
assumptio n_resoluti on	major	needs review	OCPD communications gateway	Can the OCPD communications gateway assumption be accepted for architectu re review, or must it be replaced with evidence?	Admit supplier, manufactu rer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_co mpute_co mmunicati on	none	none	none
assumptio n_resoluti on	major	needs review	ISO 15118 PLC modem	Can the ISO 15118 PLC modem assumption be accepted for architectu re review, or must it be replaced with evidence?	Admit supplier, manufactu rer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	charging_ connector _interface; control_co mpute_co mmunicati on	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	Power module stack	Can the Power module stack assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_conversion_transduction	none	none	none
assumption_resolution	major	needs review	Cooling loop assembly	Can the Cooling loop assembly assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumption_resolution	major	needs review	AC input breaker and SPD	Can the AC input breaker and SPD assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	power_distribution	none	none	none
assumption_resolution	major	needs review	MID energy meter	Can the MID energy meter assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	power_distribution; sensing_instrumentation	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	DC output contactor set	Can the DC output contactor set assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	power_distribution	none	none	none
assumption_resolution	major	needs review	Insulation monitoring device	Can the Insulation monitoring device assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation; safety_protection	none	none	none
assumption_resolution	major	needs review	Emergency stop and safety interlock set	Can the Emergency stop and safety interlock set assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	safety_protection	none	none	none
assumption_resolution	major	needs review	MID energy meter	Can the MID energy meter assumption be accepted for architecture review, or must it be replaced	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	power_distribution; sensing_instrumentation	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				with evidence?					
assumption_resolution	major	needs review	Insulation monitoring device	Can the Insulation monitoring device assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation; safety_protection	none	none	none
assumption_resolution	major	needs review	control_compute_communication -> charging_connector_interface	Can the control_compute_communication -> charging_connector_interface assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	iso15118_plc	none	none
assumption_resolution	major	needs review	control_compute_communication -> energy_conversion_transduction	Can the control_compute_communication -> energy_conversion_transduction assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	control_buses	none	none
assumption_resolution	major	needs review	energy_conversion_transduction -> charging_	Can the energy_conversion_transduction ->	Review the interface carrier, direction,	none	dc_output_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	connector_interface	charging_connector_interface assumption be accepted for architecture review, or must it be replaced with evidence?	capacity, safety margin and failure mode before detailed design or sourcing.				
			environmental_interface -> energy_conversion_transduction	Can the environmental_interface -> energy_conversion_transduction assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	coolant_loop	none	none
			power_distribution -> energy_conversion_transduction	Can the power_distribution -> energy_conversion_transduction assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	ac_input_bus	none	none
assumption_resolution	major	needs review	sensing_instrumentation -> control_compute_communication	Can the sensing_instrumentation -> control_compute_communication assumption be	Review the interface carrier, direction, capacity, safety margin and failure mode	none	metering_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				accepted for architecture review, or must it be replaced with evidence?	before detailed design or sourcing.				
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Annual energy dispensed	Can the Annual energy dispensed assumption be accepted for architecture review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performance claim.	none	none	none	none
assumption_resolution	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	DC Charger Power Level	Can the DC Charger Power Level assumption be accepted	Supply an engineering calculation, reviewer note, test plan or datasheet	power_distribution; energy_conversion_transduction; charging_connector	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for architectu re review, or must it be replaced with evidence?	-backed rationale for this envelope.	_interface; environme ntal_interf ace			
assumptio n_resoluti on	major	needs review	Power And Safety Chain	Can the Power And Safety Chain assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	power_dis tribution; energy_co nversion_t ransductio n; charging_ connector _interface; environme ntal_interf ace	none	none	none
assumptio n_resoluti on	major	needs review	Vehicle/Ba ckend Protocol Closure	Can the Vehicle/Ba ckend Protocol Closure assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	power_dis tribution; energy_co nversion_t ransductio n; charging_ connector _interface; environme ntal_interf ace	none	none	none
calculation _envelope	major	needs review	EV charger daily full-power energy	Does the EV charger daily full-power energy calculation support the architectu re envelope, or does the concept need revision?	Confirm expected sessions per day, dwell time, grid capacity, charger derating and tariff assumptions.	power_dis tribution; energy_co nversion_t ransductio n; charging_ connector _interface; environme ntal_interf ace	none	ev_charge r_daily_full _power_e nergy_kwh	none
calculation _envelope	major	ready	EV charger current at 800 V	Does the EV charger current at 800 V	Confirm output voltage range, cable	power_dis tribution; energy_co nversion_t ransductio	none	ev_charge r_800v_cu rrent_a	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				calculation support the architecture envelope, or does the concept need revision?	thermal rating, connector current limit, contactor rating and derating.	n; charging_connector_interface; environmental_interface			
interface_contract	major	ready	Control Compute Communication -> Vehicle Charging Interface via iso15118_plc	Is the iso15118_plc interface between Control Compute Communication and Vehicle Charging Interface correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_compute_communication; charging_connector_interface	iso15118_plc	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Control Compute Communication -> Power Conversion Stack via control_buses	Is the control_buses interface between Control Compute Communication and Power Conversion Stack correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_compute_communication; energy_conversion_transduction	control_buses	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Power Conversion Stack -> Vehicle Charging Interface via dc_output_bus	Is the dc_output_bus interface between Power Conversion Stack and Vehicle Charging Interface correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_conversion_transduction; charging_connector_interface	dc_output_bus	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
interface_contract	major	ready	Thermal And Environmental Interface -> Power Conversion Stack via coolant_loop	Is the coolant_loop interface between Thermal And Environmental Interface and Power Conversion Stack correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	environmental_interface; energy_conversion_transduction	coolant_loop	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	AC Input And DC Output Distribution -> Power Conversion Stack via ac_input_bus	Is the ac_input_bus interface between AC Input And DC Output Distribution and Power Conversion Stack correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	power_distribution; energy_conversion_transduction	ac_input_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Metering And Electrical Sensing -> Control Compute Communication via metering_bus	Is the metering_bus interface between Metering And Electrical Sensing and Control Compute Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	sensing_instrumentation; control_compute_communication	metering_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
module_location	major	ready	Vehicle Charging Interface	Does Vehicle Charging Interface have the right	Named engineering reviewer sign-off, change request or	charging_connector_interface	dc_output_bus; ccs2_vehicle_interface; iso15118_	evcharger_power_level; evcharger_power_safety_chain;	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				responsibi lity boundary, submodul e split and interface ownership for this concept?	deferral note against the module allocation.		plc; control_pil ot; protective _earth; coolant_lo op; service_ac cess; control_bu s	evcharger _protocol_ closure	
module_all ocation	major	ready	Control Compute Communication	Does Control Compute Communication have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	control_co mpute_co mmunicati on	control_bu s; iso15118_ plc; ocpp_net work; metering_ bus; alarm_bus ; service_p ort	none	none
module_all ocation	major	ready	Power Conversio n Stack	Does Power Conversio n Stack have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	energy_co nversion_t ransductio n	ac_input_ bus; dc_output _bus; coolant_lo op; control_bu s; protective _earth; sensor_bu s	evcharger _power_le vel; evcharger _power_sa fety_chain; evcharger _protocol_ closure	none
module_all ocation	major	ready	Thermal And Environme ntal Interface	Does Thermal And Environme ntal Interface have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	environme ntal_interf ace	coolant_lo op; airflow_pa th; condensat e_drain; service_ac cess; protective _earth; sensor_bu s	evcharger _power_le vel; evcharger _power_sa fety_chain; evcharger _protocol_ closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	User Payment And HMI	Does User Payment And HMI have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	hmi_ergonomics	ocpp_network; control_bu s; ccs2_vehicle_interface; service_access	none	none
module_all ocation	major	ready	Maintenance And Commissioning	Does Maintenance And Commissioning have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	maintenance_serviceability	service_access; service_port; ocpp_network; metering_bus; coolant_loop; control_bu s	none	none
module_all ocation	major	ready	AC Input And DC Output Distribution	Does AC Input And DC Output Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_distribution	ac_input_bus; dc_output_bus; protective_earth; metering_bus; service_access; control_bu s	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
module_all ocation	major	ready	Safety Protection	Does Safety Protection have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	safety_protection	emergency_stop; insulation_monitoring ; protective_earth; alarm_bus ; dc_output_bus; control_bu s; ccs2_vehicle_interface	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Metering And Electrical Sensing	Does Metering And Electrical Sensing have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	sensing_in strumenta tion	metering_ bus; dc_output _bus; insulation_ monitoring ; sensor_bu s; control_bu s; alarm_bus ; ccs2_vehi cle_interfa ce; control_pil ot	none	none
module_all ocation	major	ready	Cabinet Structure And Containme nt	Does Cabinet Structure And Containme nt have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	mechanica l_mounts; service_ac cess; protective _earth; ccs2_vehi cle_interfa ce	none	none
submodul e_allocatio n	major	ready	Vehicle Charging Interface / CCS2 cable connector	Do the componen t candidate s and interfaces assigned to CCS2 cable connector make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	charging_ connector _interface	dc_output _bus; ccs2_vehi cle_interfa ce; control_pil ot; protective _earth	evcharger _power_le vel; evcharger _power_sa fety_chain; evcharger _protocol_ closure	none
submodul e_allocatio n	major	ready	Vehicle Charging Interface / Connector cooling coupling	Do the componen t candidate s and interfaces assigned to Connector cooling coupling	Reviewer note confirming the submodul e role, componen t allocation and any missing	charging_ connector _interface	coolant_lo op; ccs2_vehi cle_interfa ce; service_ac cess	evcharger _power_le vel; evcharger _power_sa fety_chain; evcharger _protocol_ closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				make engineering sense for its stated purpose?	carriers or capacity checks.				
submodule_allocation	major	ready	Vehicle Charging Interface / Vehicle communication frontend	Do the component candidate s and interfaces assigned to Vehicle communication frontend make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	charging_connector_interface	iso15118_plc; control_pilot; control_buss	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
submodule_allocation	major	ready	Control Compute Communication / Backend gateway	Do the component candidate s and interfaces assigned to Backend gateway make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	ocpp_network; service_port	none	none
submodule_allocation	major	ready	Control Compute Communication / Charge controller	Do the component candidate s and interfaces assigned to Charge controller make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	control_buss; metering_bus; alarm_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Control Compute Communication / Vehicle protocol stack	Do the component candidates and interfaces assigned to Vehicle protocol stack make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	iso15118_plc; control_buses; metering_bus	none	none
submodule_allocation	major	ready	Power Conversion Stack / Converter supervision	Do the component candidates and interfaces assigned to Converter supervision make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	control_buses; sensor_buses; coolant_loop	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
submodule_allocation	major	ready	Power Conversion Stack / Output filtering	Do the component candidates and interfaces assigned to Output filtering make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	dc_output_bus; protective_earth	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
submodule_allocation	major	ready	Power Conversion Stack / Rectifier power modules	Do the component candidates and interfaces assigned to Rectifier power modules make engineering sense	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	ac_input_bus; dc_output_bus; coolant_loop; control_buses	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for its stated purpose?					
submodule_allocation	major	ready	Thermal And Environmental Interface / Liquid cooling loop	Do the component candidates and interfaces assigned to Liquid cooling loop make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	coolant_loop; sensor_buses; service_access	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
submodule_allocation	major	ready	Thermal And Environmental Interface / Thermal air path	Do the component candidates and interfaces assigned to Thermal air path make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	airflow_path; coolant_loop; protective_earth	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
submodule_allocation	major	ready	Thermal And Environmental Interface / Weatherproofing	Do the component candidates and interfaces assigned to Weatherproofing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	condensate_drain; service_access	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
submodule_allocation	major	ready	User Payment And HMI / Payment and identity	Do the component candidates and interfaces assigned to Payment and identity make	Reviewer note confirming the submodule role, component allocation and any missing carriers or	hmi_ergonomics	ocpp_network; control_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				engineering sense for its stated purpose?	capacity checks.				
submodule_allocation	major	ready	User Payment And HMI / User cable guidance	Do the component candidate s and interfaces assigned to User cable guidance make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	ccs2_vehicle_interface; service_access	none	none
submodule_allocation	major	ready	User Payment And HMI / User display panel	Do the component candidate s and interfaces assigned to User display panel make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	control_buss; ocpp_network	none	none
submodule_allocation	major	ready	Maintenance And Commissioning / Commissioning access	Do the component candidate s and interfaces assigned to Commissioning access make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_serviceability	service_access; service_port; metering_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodul e_allocation	major	ready	Maintenan ce And Commissi oning / Remote diagnostic s	Do the componen t candidate s and interfaces assigned to Remote diagnostic s make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	maintenan ce_service ability	ocpp_net work; service_p ort; control_bu s	none	none
submodul e_allocation	major	ready	Maintenan ce And Commissi oning / Replaceable modules	Do the componen t candidate s and interfaces assigned to Replaceable modules make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	maintenan ce_service ability	service_ac cess; control_bu s; coolant_lo op	none	none
submodul e_allocation	major	ready	AC Input And DC Output Distributio n / AC input switchgea r	Do the componen t candidate s and interfaces assigned to AC input switchgea r make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	power_dis tribution	ac_input_ bus; protective _earth; service_ac cess	evcharger _power_le vel; evcharger _power_sa fety_chain; evcharger _protocol_ closure	none
submodul e_allocation	major	ready	AC Input And DC Output Distributio n / Billing metering branch	Do the componen t candidate s and interfaces assigned to Billing metering branch make engineerin g sense for its	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	power_dis tribution	metering_ bus; dc_output _bus; service_ac cess	evcharger _power_le vel; evcharger _power_sa fety_chain; evcharger _protocol_ closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				stated purpose?					
submodule_allocation	major	ready	AC Input And DC Output Distribution / DC output switchgear	Do the component candidates and interfaces assigned to DC output switchgear make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	dc_output_bus; control_buses; protective_earth	evcharger_power_level; evcharger_power_safety_chain; evcharger_protocol_closure	none
submodule_allocation	major	ready	Safety Protection / DC fault protection	Do the component candidates and interfaces assigned to DC fault protection make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	dc_output_bus; insulation_monitoring; alarm_bus	none	none
submodule_allocation	major	ready	Safety Protection / Emergency stop chain	Do the component candidates and interfaces assigned to Emergency stop chain make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	emergency_stop; alarm_bus; control_buses	none	none

Showing first 70 of 77 engineering review questions; export CSV for the full pack.

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
DC Charger Power Level	pass	150 kW	50 to 350 kW for common public DC fast-charger classes; >350 kW needs heavier cable, grid and cooling evidence	150 kW is a plausible DC fast-charger output before detailed grid, cable and thermal derating evidence.
Vehicle/Backend Protocol Closure	pass	present	CCS2, ISO 15118 PLC, OCPP network and metering bus present	Vehicle handshake, backend session state and billable metering are connected before sourcing.
Power And Safety Chain	pass	present	AC input, DC output, cooling, insulation monitoring, emergency stop and protective earth present	High-power conversion and user/service safety interfaces are explicitly allocated.

Engineering Calculation Ledger

2 calculations: 1 within deterministic envelope, 1 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
EV charger current at 800 V	within envelope	(dc_power_kw * 1,000) / 800	187.5 A	<=500 A is a broad liquid-cooled CCS concept envelope before connector and cable proof.	187.5 A at an assumed 800 V output point.	Confirm output voltage range, cable thermal rating, connector current limit, contactor rating and derating.
EV charger daily full-power energy	needs review	dc_power_kw * 4	600 kWh/day	Utilisation assumption only; revenue and thermal duty need site-specific evidence.	600 kWh/day assumes 4 equivalent full-power hours.	Confirm expected sessions per day, dwell time, grid capacity, charger derating and tariff assumptions.

Engineering Assumption Ledger

30 assumptions: 1 brief-supported, 6 model-present, 8 needing engineering review, 15 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 15.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
DC output power	brief_requirement	brief supported	no	no	DC output power = 150 kW is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Annual energy dispensed	derived_metric	review required	no	no	Annual energy dispensed uses dc_power_kw * equivalent_full_power_hours_per_day * days_per_year with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
DC Charger Power Level	sanity_envelope	review required	no	no	DC Charger Power Level currently evaluates as pass against 50 to 350 kW for common public DC fast-charger classes; >350 kW needs heavier cable, grid and cooling evidence.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Vehicle/Backend Protocol Closure	sanity_envelope	review required	no	no	Vehicle/Backend Protocol Closure currently evaluates as pass against CCS2, ISO 15118 PLC, OCPP network and metering bus present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Power And Safety Chain	sanity_envelope	review required	no	no	Power And Safety Chain currently evaluates as pass against AC input, DC	Supply an engineering calculation, reviewer note, test plan or datasheet-

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					output, cooling, insulation monitoring, emergency stop and protective earth present.	backed rationale for this envelope.
power_distribution -> energy_conversion_transduction	interface_closure	model present	no	no	ac_input_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
energy_conversion_transduction -> charging_connector_interface	interface_closure	model present	no	no	dc_output_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_computer_communication -> charging_connector_interface	interface_closure	model present	no	no	iso15118_plc is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_computer_communication -> energy_conversion_transduction	interface_closure	model present	no	no	control_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
sensing_instrumentation -> control_computer_communication	interface_closure	model present	no	no	metering_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
environmental_interface -> energy_conversion_transduction	interface_closure	model present	no	no	coolant_loop is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
						design or sourcing.
AC input breaker and SPD	critical_component	source required	no	yes	AC input breaker and SPD is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
DC output contactor set	critical_component	source required	no	yes	DC output contactor set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
MID energy meter	critical_component	source required	no	yes	MID energy meter is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Power module stack	critical_component	source required	no	yes	Power module stack is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
CCS2 liquid-cooled cable and connector	critical_component	source required	no	yes	CCS2 liquid-cooled cable and connector is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
ISO 15118 PLC modem	critical_component	source required	no	yes	ISO 15118 PLC modem is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
OCPP communications gateway	critical_component	source required	no	yes	OCPP communications gateway is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
ISO 15118 PLC modem	critical_component	source required	no	yes	ISO 15118 PLC modem is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
MID energy meter	critical_component	source required	no	yes	MID energy meter is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Insulation monitoring device	critical_component	source required	no	yes	Insulation monitoring device is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Cooling loop assembly	critical_component	source required	no	yes	Cooling loop assembly is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Emergency stop and safety interlock set	critical_component	source required	no	yes	Emergency stop and safety interlock set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Insulation monitoring device	critical_component	source required	no	yes	Insulation monitoring device is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
IEC-61851	compliance_evidence	review required	no	no	Conductive charging system requirements is relevant to this product class in UK/EU.	Charging modes, control pilot and safety-function evidence.
IEC-62196-3	compliance_evidence	review required	no	no	DC vehicle connector requirements is relevant to this product class in UK/EU.	CCS2 connector rating, cable assembly and temperature evidence.
ISO-15118	compliance_evidence	review required	no	no	Vehicle-to-grid communication interface is relevant to this product class in Global / EU.	PLC communication, Plug and Charge and session-state evidence where applicable.
OCPP-1.6J/2.0.1	compliance_evidence	review required	no	no	Charge point backend protocol is relevant to this product class in Operator requirement.	Backend session, metering, fault and firmware-management evidence.

Engineering Verification Plan

34 activities: 20 ready for review, 7 open, 7 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review AC Input And DC Output Distribution architecture allocation	ready	design_review	AC Input And DC Output Distribution	dc_power_kw	ac_input_buss; dc_output_bus; protective_earth; metering_buss; service_access; control_bus	ac_input_breaker_and_spd; lockable_ac_isolator; surge_arrestor_cartridge; input_terminal_shroud; dc_output_contactor_set; precharge_resistor_module; dc_output_fuse_carrier; output_polarity_label; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify AC Input And DC Output Distribution requirement calculations	ready	calculation	AC Input And DC Output Distribution	dc_power_kw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for AC Input And DC Output Distribution critical components	blocked	source_evidence	AC Input And DC Output Distribution	dc_power_kw	none	ac_input_breaker_and_spd; dc_output_contactor_set; mid_energy_meter	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	ac_input_breaker_and_spd dc_output_contactor_set mid_energy_meter

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Power Conversion Stack architecture allocation	ready	design_review	Power Conversion Stack	dc_power_kw	ac_input_buses; dc_output_bus; coolant_loop; control_bus; protective_earth; sensor_bus	power_module_stack; power_module_backplane; module_location_rail; dc_link_capacitor_bank; dc_output_choke; emi_filter_assembly; snubber_network_board; thermal_interface_pad_set; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Power Conversion Stack requirement calculations	ready	calculation	Power Conversion Stack	dc_power_kw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Power Conversion Stack critical components	blocked	source_evidence	Power Conversion Stack	dc_power_kw	none	power_module_stack	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	power_module_stack
Review Vehicle Charging Interface architecture allocation	ready	design_review	Vehicle Charging Interface	dc_power_kw	dc_output_bus; ccs2_vehicle_interface; iso15118_plc; control_pilot; protective_earth; coolant_loop; service_access; control_bus	ccs2_liquid_cooled_cable_and_connector; connector_temperature_sensor_pair; proximity_pilot_resistor_set; cable_strain_relief_boot; liquid_cooled_cable_manifold; quick_connect_coolant_coupler; coolant_leak_catch_tray; cable_coolant_temperature_probe; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Vehicle Charging Interface requirement calculations	ready	calculation	Vehicle Charging Interface	dc_power_kw	none	none	Each linked requirement has a calculation, sanity check or	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Vehicle Charging Interface critical components	blocked	source_evidence	Vehicle Charging Interface	dc_power_kw	none	ccs2_liquid_cooled_cable_and_connector; iso_15118_plc_modem	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	ccs2_liquid_cooled_cable_and_connector iso_15118_plc_modem
Review Control Compute Communication architecture allocation	ready	design_review	Control Compute Communication	none	control_bus; iso15118_plc; ocpp_network; metering_bus; alarm_bus; service_port	charge_controller_pcb; safety_io_terminal_block; real_time_clock_module; nonvolatile_event_log_memory; ocpp_communications_gateway; lte_router_module; ethernet_surge_protector;	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						secure_element_module; plus 4 more	interface. No blocker or major architecture issue is attached to the module.	
Collect source-backed evidence for Control Compute Communication critical components	blocked	source_evidence	Control Compute Communication	none	none	ocpp_communications_gateway; iso_15118_plc_modem	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	ocpp_communications_gateway iso_15118_plc_modem
Review Metering And Electrical Sensing architecture allocation	ready	design_review	Metering And Electrical Sensing	none	metering_bus; dc_output_bus; insulation_monitoring; sensor_bus; control_bus; alarm_bus; ccs2_vehicle_interface; control_pilot	mid_energy_meter; dc_voltage_transducer; dc_current_shunt; meter_pulse_output_isolator; insulation_monitoring_device; imd_coupling_network; fault_status_relay; test_resistor_plug; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Metering And Electrical Sensing critical components	blocked	source_evidence	Metering And Electrical Sensing	none	none	mid_energy_meter; insulation_monitoring_device	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	mid_energy_meter insulation_monitoring_device
Review Thermal And Environmental Interface architecture allocation	ready	design_review	Thermal And Environmental Interface	dc_power_kw	coolant_loop; airflow_path; condensate_drain; service_access; protective_earth; sensor_bus	cooling_loop_assembly; coolant_pump; plate_radiator; coolant_reservoir; cabinet_fan_tray; inlet_filter_mat; exhaust_louvre_assembly; airflow_proving_switch; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Thermal And Environmental Interface requirement calculations	ready	calculation	Thermal And Environmental Interface	dc_power_kw	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Thermal And Environmental Interface critical components	blocked	source_evidence	Thermal And Environmental Interface	dc_power_kw	none	cooling_loop_assembly	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	cooling_loop_assembly
Review Cabinet Structure And Containment architecture allocation	ready	design_review	Cabinet Structure And Containment	none	mechanical_mounts; service_access; protective_earth; ccs2_vehicle_interface	powder_coated_steel_cabinet; front_service_door; anti_vandal_hinge_set; plinth_mounting_kit; connector_holster; cable_retractor_arm; cable_bend_limiter; parking_status_flag; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							No blocker or major architecture issue is attached to the module.	
Review Safety Protection architecture allocation	ready	design_review	Safety Protection	none	emergency_stop; insulation_monitoring; protective_earth; alarm_bus; dc_output_bus; control_bus; ; ccs2_vehicle_interface	emergency_stop_and_safety_interlock_set; e_stop_mushroom_button; door_interlock_switch; safety_relay_module; insulation_monitoring_device; dc_arc_detection_board; ; output_contactor_weld_check_circuit; fault_discharge_resistor; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Collect source-backed evidence for Safety Protection critical components	blocked	source_evidence	Safety Protection	none	none	emergency_stop_and_safety_interlock_set; insulation_monitoring_device	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	emergency_stop_and_safety_interlock_set insulation_monitoring_device

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Maintenance And Commissioning architecture allocation	ready	design_review	Maintenance And Commissioning	none	service_access; service_port; ocpp_network; metering_buses; coolant_loop; control_bus	commissioning_terminal_block; meter_test_port; output_test_socket_converter; commissioning_checklist_card; slide_out_power_module_rail; quick_disconnect_coolant_fitting; gateway_din_rail_carrier; ; spare_fuse_and_link_kit; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review User Payment And HMI architecture allocation	ready	design_review	User Payment And HMI	none	ocpp_network; control_bus; ; ccs2_vehicle_interface; service_access	touchscreen_hmi_panel; status_led_light_bar; audio_buzzer; sunlight_readable_display_cover; rfid_reader_module; contactless_payment_terminal; receipt_qr_code_generator; privacy_label; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify ac_input_buses interface contract	ready	interface_review	AC Input And DC Output Distribution -> Power Conversion Stack	none	ac_input_buses	ac_input_breaker_and_spd; lockable_ac_isolator; surge_arrestor_cartridge; input_terminal_shroud;	Both endpoint modules declare the interface. At least one submodule carrier is named on each	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						power_module_stack; power_module_backplane; module_location_rail; dc_link_capacitor_bank	endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Verify dc_output_bus interface contract	ready	interface_review	Power Conversion Stack -> Vehicle Charging Interface	none	dc_output_bus	power_module_stack; power_module_backplane; module_location_rail; dc_link_capacitor_bank; dc_output_choke; emi_filter_assembly; snubber_network_board; thermal_interface_pad_set; plus 4 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify iso15118_plc interface contract	ready	interface_review	Control Compute Communication -> Vehicle Charging Interface	none	iso15118_plc	iso_15118_plc_modem; plug_and_charge_certificate_store; charging_session_firmware_image; protocol_debug_header; iso_15118_plc_modem; control_pilot_interface_board; proximity_detection_input; vehicle_comms_isolation_transformer	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify control_bus interface contract	ready	interface_review	Control Compute Communication -> Power	none	control_bus	charge_controller_pcb; safety_io_terminal_block;	Both endpoint modules declare the interface. At	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
			Conversion Stack			real_time_clock_module ; nonvolatile_event_log_memory; iso_15118_plc_modem; plug_and_charge_certificate_store; charging_session_firmware_image; protocol_debug_header ; plus 8 more	least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Verify metering_bus interface contract	ready	interface_review	Metering And Electrical Sensing -> Control Compute Communication	none	metering_bus	mid_energy_meter; dc_voltage_transducer; dc_current_shunt; meter_pulse_output_isolator; charge_controller_pcb; safety_io_terminal_block; real_time_clock_module ; nonvolatile_event_log_memory; plus 4 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify coolant_loop interface contract	ready	interface_review	Thermal And Environmental Interface -> Power Conversion Stack	none	coolant_loop	cooling_loop_assembly ; coolant_pump; plate_radiator; coolant_reservoir; cabinet_fan_tray; inlet_filter_mat; exhaust_louvre_assembly; airflow_proving_switch; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Plan evidence for IEC-61851	open	compliance_review	System Compliance	none	none	none	Charging modes, control pilot and safety-function evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for IEC-62196-3	open	compliance_review	System Compliance	none	none	none	CCS2 connector rating, cable assembly and temperature evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for ISO-15118	open	compliance_review	System Compliance	none	none	none	PLC communication, Plug and Charge and session-state evidence where applicable.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for OCPP-1.6J/2.0.1	open	compliance_review	System Compliance	none	none	none	Backend session, metering, fault and firmware-management evidence.	Compliance evidence not yet collected in scratch prototype.
Review risk control for DC output fault or insulation failure	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for Connector overheating at high current	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by	Risk evidence not yet reviewed in scratch prototype.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	
Review risk control for Backend or payment outage stops sessions	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 27 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review AC Input And DC Output Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify AC Input And DC Output Distribution requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for AC Input And DC Output Distribution critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Power Conversion Stack architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Power Conversion Stack requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Power Conversion Stack critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Vehicle Charging Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Vehicle Charging Interface requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Vehicle Charging Interface critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Control Compute Communication architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Control Compute Communication critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Metering And Electrical Sensing	pending	design_review			Collect reviewer evidence through

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
architecture allocation					the verification intake template.
Collect source-backed evidence for Metering And Electrical Sensing critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Thermal And Environmental Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Thermal And Environmental Interface requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Thermal And Environmental Interface critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Cabinet Structure And Containment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Safety Protection architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Safety Protection critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Maintenance And Commissioning architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review User Payment And HMI architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify ac_input_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify dc_output_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify iso15118_plc interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify control_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify metering_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify coolant_loop interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for IEC-61851	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for IEC-62196-3	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for ISO-15118	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for OCPP-1.6J/2.0.1	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for DC output fault or insulation failure	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Connector overheating at high current	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Backend or payment outage stops sessions	pending	compliance_review			Collect reviewer evidence through the verification intake template.

AC Input And DC Output Distribution

Protects the incoming supply and DC output path for a roughly 150 kW charger.

ac_input_bus	dc_output_bus	protective_earth	metering_bus	service_access
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AC input switchgear

Protects and isolates the grid connection before conversion.

Components: AC input breaker and SPD, lockable AC isolator, surge arrestor cartridge, input terminal shroud

DC output switchgear

Connects and isolates the charging output path.

Components: DC output contactor set, precharge resistor module, DC output fuse carrier, output polarity label

Billing metering branch

Measures delivered energy and exposes metering data to control.

Components: MID energy meter, meter CT shunt set, meter seal kit, meter calibration label

Power Conversion Stack

Converts protected AC input into controlled high-power DC output.

ac_input_bus	dc_output_bus	coolant_loop	control_bus	protective_earth
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Rectifier power modules

Performs modular AC/DC conversion.

Components: Power module stack, power module backplane, module locking rail, DC-link capacitor bank

Output filtering

Reduces ripple and conducted emissions on charging output.

Components: DC output choke, EMI filter assembly, snubber network board, thermal interface pad set

Converter supervision

Monitors module health and coordinates derating.

Components: module health monitor board, heatsink temperature probe, fan/coolant derating table, firmware recovery jumper

Vehicle Charging Interface

Presents the CCS2 conductive and communication interface to the vehicle.

dc_output_bus	ccs2_vehicle_interface	iso15118_plc	control_pilot	protective_earth
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CCS2 cable connector

Routes high-current DC and proximity/pilot connections.

Components: CCS2 liquid-cooled cable and connector, connector temperature sensor pair, proximity pilot resistor set, cable strain relief boot

Connector cooling coupling

Connects high-current cable cooling to the charger thermal loop.

Components: liquid-cooled cable manifold, quick-connect coolant coupler, coolant leak catch tray, cable coolant temperature probe

Vehicle communication frontend

Handles control pilot and PLC communication with the vehicle.

Components: ISO 15118 PLC modem, control pilot interface board, proximity detection input, vehicle comms isolation transformer

Control Compute Communication

Coordinates charge sessions, power-module commands, backend state and safety reactions.

- control_bus
- iso15118_plc
- ocpp_network
- metering_bus
- alarm_bus
- service_port

Charge controller

Runs session state, output limits and fault transitions.

Components: charge controller PCB, safety IO terminal block, real-time clock module, nonvolatile event log memory

Backend gateway

Connects the charge point to operator systems.

Components: OCPP communications gateway, LTE router module, ethernet surge protector, secure element module

Vehicle protocol stack

Coordinates ISO 15118, cable limits and metering sessions.

Components: ISO 15118 PLC modem, Plug and Charge certificate store, charging session firmware image, protocol debug header

Metering And Electrical Sensing

Measures billing energy, voltage, current, insulation and connector state.

- metering_bus
- dc_output_bus
- insulation_monitoring
- sensor_bus
- control_bus

Energy metering

Measures energy delivered to the vehicle.

Components: MID energy meter, DC voltage transducer, DC current shunt, meter pulse output isolator

Insulation fault detection

Detects loss of isolation on the DC charging path.

Components: Insulation monitoring device, IMD coupling network, fault status relay, test resistor plug

Connector sensing

Monitors cable, latch and temperature states.

Components: connector latch microswitch, handle temperature sensor, cable identification resistor reader, holster presence switch

Thermal And Environmental Interface

Rejects power-electronics and cable heat while protecting outdoor electronics.

- coolant_loop
- airflow_path
- condensate_drain
- service_access
- protective_earth

Liquid cooling loop

Circulates coolant through power modules and liquid-cooled cable.

Components: Cooling loop assembly, coolant pump, plate radiator, coolant reservoir

Thermal air path

Moves air through cabinet heat exchangers and filters.

Components: cabinet fan tray, inlet filter mat, exhaust louvre assembly, airflow proving switch

Weatherproofing

Protects the charger from rain, condensation and UV exposure.

Components: cabinet drain grommet, door gasket set, anti-condensation heater, UV-rated cable cover

Cabinet Structure And Containment

Houses high-power electronics, cable management and outdoor installation features.

- mechanical_mounts
- service_access
- protective_earth
- ccs2_vehicle_interface

Outdoor cabinet

Carries power electronics and protects against weather and impact.

Components: powder-coated steel cabinet, front service door, anti-vandal hinge set, plinth mounting kit

Cable holster and boom

Manages heavy high-current cable during user operation.

Components: connector holster, cable retractor arm, cable bend limiter, parking status flag

Internal compartmentation

Separates AC input, DC output, control and user zones.

Components: segregated AC/DC barrier, control compartment shield, service warning placard, earthing braid set

Safety Protection

Protects users, vehicles and service personnel from high-power charging hazards.

- emergency_stop
- insulation_monitoring
- protective_earth
- alarm_bus
- dc_output_bus

Emergency stop chain

Trips output contactors and records user/service stop events.

Components: Emergency stop and safety interlock set, E-stop mushroom button, door interlock switch, safety relay module

DC fault protection

Detects and interrupts DC-side electrical faults.

Components: Insulation monitoring device, DC arc detection board, output contactor weld check circuit, fault discharge resistor

User access protection

Prevents user contact with live parts and communicates safe operation.

Components: touch-safe connector shutter, protective earth continuity strap, charging status beacon, safety instruction label

Maintenance And Commissioning

Supports installation, commissioning, remote diagnosis and field replacement.

- service_access
- service_port
- ocpp_network
- metering_bus
- coolant_loop

Commissioning access

Exposes safe setup and verification points.

Components: commissioning terminal block, meter test port, output test socket cover, commissioning checklist card

Replaceable modules

Makes high-failure assemblies field swappable.

Components: slide-out power module rail, quick-disconnect coolant fitting, gateway DIN rail carrier, spare fuse and link kit

Remote diagnostics

Collects and exports health data for service teams.

Components: diagnostic log exporter, remote firmware update agent, health telemetry packet schema, service VPN profile

User Payment And HMI

Guides charging sessions, payment, status and accessibility at the charger front end.

ocpp_network	control_bus	ccs2_vehicle_interface	service_access
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User display panel

Presents session state, price, energy and fault messages.

Components: touchscreen HMI panel, status LED light bar, audio buzzer, sunlight-readable display cover

Payment and identity

Supports card, RFID or app-based authorisation.

Components: RFID reader module, contactless payment terminal, receipt QR code generator, privacy label

User cable guidance

Helps users lift, park and confirm the heavy CCS2 connector.

Components: connector instruction graphic, cable park indicator, accessibility reach label, session stop button

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: **£0**

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
AC input breaker and SPD	1	candidate	unpriced	unpriced				
lockable AC isolator	1	candidate	unpriced	unpriced				
surge arrestor cartridge	1	candidate	unpriced	unpriced				
input terminal shroud	1	candidate	unpriced	unpriced				
DC output contactor set	1	candidate	unpriced	unpriced				
precharge resistor module	1	candidate	unpriced	unpriced				
DC output fuse carrier	1	candidate	unpriced	unpriced				
output polarity label	1	candidate	unpriced	unpriced				
MID energy meter	1	candidate	unpriced	unpriced				
meter CT shunt set	1	candidate	unpriced	unpriced				
meter seal kit	1	candidate	unpriced	unpriced				
meter calibration label	1	candidate	unpriced	unpriced				
Power module stack	1	candidate	unpriced	unpriced				
power module backplane	1	candidate	unpriced	unpriced				
module locking rail	1	candidate	unpriced	unpriced				
DC-link capacitor bank	1	candidate	unpriced	unpriced				
DC output choke	1	candidate	unpriced	unpriced				
EMI filter assembly	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
snubber network board	1	candidate	unpriced	unpriced				
thermal interface pad set	1	candidate	unpriced	unpriced				
module health monitor board	1	candidate	unpriced	unpriced				
heatsink temperature probe	1	candidate	unpriced	unpriced				
fan/coolant derating table	1	candidate	unpriced	unpriced				
firmware recovery jumper	1	candidate	unpriced	unpriced				
CCS2 liquid- cooled cable and connector	1	candidate	unpriced	unpriced				
connector temperature sensor pair	1	candidate	unpriced	unpriced				
proximity pilot resistor set	1	candidate	unpriced	unpriced				
cable strain relief boot	1	candidate	unpriced	unpriced				
liquid- cooled cable manifold	1	candidate	unpriced	unpriced				
quick- connect coolant coupler	1	candidate	unpriced	unpriced				
coolant leak catch tray	1	candidate	unpriced	unpriced				
cable coolant temperature probe	1	candidate	unpriced	unpriced				
ISO 15118 PLC modem	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
control pilot interface board	1	candidate	unpriced	unpriced				
proximity detection input	1	candidate	unpriced	unpriced				
vehicle comms isolation transformer	1	candidate	unpriced	unpriced				
charge controller PCB	1	candidate	unpriced	unpriced				
safety IO terminal block	1	candidate	unpriced	unpriced				
real-time clock module	1	candidate	unpriced	unpriced				
nonvolatile event log memory	1	candidate	unpriced	unpriced				
OCPP communications gateway	1	candidate	unpriced	unpriced				
LTE router module	1	candidate	unpriced	unpriced				
ethernet surge protector	1	candidate	unpriced	unpriced				
secure element module	1	candidate	unpriced	unpriced				
ISO 15118 PLC modem	1	candidate	unpriced	unpriced				
Plug and Charge certificate store	1	candidate	unpriced	unpriced				
charging session firmware image	1	candidate	unpriced	unpriced				
protocol debug header	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
MID energy meter	1	candidate	unpriced	unpriced				
DC voltage transducer	1	candidate	unpriced	unpriced				
DC current shunt	1	candidate	unpriced	unpriced				
meter pulse output isolator	1	candidate	unpriced	unpriced				
Insulation monitoring device	1	candidate	unpriced	unpriced				
IMD coupling network	1	candidate	unpriced	unpriced				
fault status relay	1	candidate	unpriced	unpriced				
test resistor plug	1	candidate	unpriced	unpriced				
connector latch microswitch	1	candidate	unpriced	unpriced				
handle temperature sensor	1	candidate	unpriced	unpriced				
cable identification resistor reader	1	candidate	unpriced	unpriced				
holster presence switch	1	candidate	unpriced	unpriced				
Cooling loop assembly	1	candidate	unpriced	unpriced				
coolant pump	1	candidate	unpriced	unpriced				
plate radiator	1	candidate	unpriced	unpriced				
coolant reservoir	1	candidate	unpriced	unpriced				
cabinet fan tray	1	candidate	unpriced	unpriced				
inlet filter mat	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
exhaust louvre assembly	1	candidate	unpriced	unpriced				
airflow proving switch	1	candidate	unpriced	unpriced				
cabinet drain grommet	1	candidate	unpriced	unpriced				
door gasket set	1	candidate	unpriced	unpriced				
anti-condensation heater	1	candidate	unpriced	unpriced				
UV-rated cable cover	1	candidate	unpriced	unpriced				
powder-coated steel cabinet	1	candidate	unpriced	unpriced				
front service door	1	candidate	unpriced	unpriced				
anti-vandal hinge set	1	candidate	unpriced	unpriced				
plinth mounting kit	1	candidate	unpriced	unpriced				
connector holster	1	candidate	unpriced	unpriced				
cable retractor arm	1	candidate	unpriced	unpriced				
cable bend limiter	1	candidate	unpriced	unpriced				
parking status flag	1	candidate	unpriced	unpriced				
segregated AC/DC barrier	1	candidate	unpriced	unpriced				
control compartment shield	1	candidate	unpriced	unpriced				
service warning placard	1	candidate	unpriced	unpriced				
earthing braid set	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Emergency stop and safety interlock set	1	candidate	unpriced	unpriced				
E-stop mushroom button	1	candidate	unpriced	unpriced				
door interlock switch	1	candidate	unpriced	unpriced				
safety relay module	1	candidate	unpriced	unpriced				
Insulation monitoring device	1	candidate	unpriced	unpriced				
DC arc detection board	1	candidate	unpriced	unpriced				
output contactor weld check circuit	1	candidate	unpriced	unpriced				
fault discharge resistor	1	candidate	unpriced	unpriced				
touch-safe connector shutter	1	candidate	unpriced	unpriced				
protective earth continuity strap	1	candidate	unpriced	unpriced				
charging status beacon	1	candidate	unpriced	unpriced				
safety instruction label	1	candidate	unpriced	unpriced				
commissioning terminal block	1	candidate	unpriced	unpriced				
meter test port	1	candidate	unpriced	unpriced				
output test socket cover	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
commissioning checklist card	1	candidate	unpriced	unpriced				
slide-out power module rail	1	candidate	unpriced	unpriced				
quick-disconnect coolant fitting	1	candidate	unpriced	unpriced				
gateway DIN rail carrier	1	candidate	unpriced	unpriced				
spare fuse and link kit	1	candidate	unpriced	unpriced				
diagnostic log exporter	1	candidate	unpriced	unpriced				
remote firmware update agent	1	candidate	unpriced	unpriced				
health telemetry packet schema	1	candidate	unpriced	unpriced				
service VPN profile	1	candidate	unpriced	unpriced				
touchscreen HMI panel	1	candidate	unpriced	unpriced				
status LED light bar	1	candidate	unpriced	unpriced				
audio buzzer	1	candidate	unpriced	unpriced				
sunlight-readable display cover	1	candidate	unpriced	unpriced				
RFID reader module	1	candidate	unpriced	unpriced				
contactless payment terminal	1	candidate	unpriced	unpriced				
receipt QR code generator	1	candidate	unpriced	unpriced				
privacy label	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
connector instruction graphic	1	candidate	unpriced	unpriced				
cable park indicator	1	candidate	unpriced	unpriced				
accessibility reach label	1	candidate	unpriced	unpriced				
session stop button	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	120	0	120	13

Critical Sourcing Worklist

Component	Qty	Reason
AC input breaker and SPD	1 candidate	Required class-critical line must be sourced before BoM can pass.
DC output contactor set	1 candidate	Required class-critical line must be sourced before BoM can pass.
MID energy meter	1 candidate	Required class-critical line must be sourced before BoM can pass.
Power module stack	1 candidate	Required class-critical line must be sourced before BoM can pass.
CCS2 liquid-cooled cable and connector	1 candidate	Required class-critical line must be sourced before BoM can pass.
ISO 15118 PLC modem	1 candidate	Required class-critical line must be sourced before BoM can pass.
OCPP communications gateway	1 candidate	Required class-critical line must be sourced before BoM can pass.
ISO 15118 PLC modem	1 candidate	Required class-critical line must be sourced before BoM can pass.
MID energy meter	1 candidate	Required class-critical line must be sourced before BoM can pass.
Insulation monitoring device	1 candidate	Required class-critical line must be sourced before BoM can pass.
Cooling loop assembly	1 candidate	Required class-critical line must be sourced before BoM can pass.
Emergency stop and safety interlock set	1 candidate	Required class-critical line must be sourced before BoM can pass.
Insulation monitoring device	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/120 priced lines admitted. Critical coverage 0/13. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: 3 canonical component IDs appear across 6 allocated lines. Resolve these before sourcing or deduplicating costs.

Component ID	Allocated Lines	Critical Lines	Description	Total Quantity	Review Action
insulation_monitoring_device	2	2	Insulation monitoring device	2 candidate	Resolve whether these allocations are one shared physical item or separate install locations before sourcing, costing or deduplicating.
iso_15118_plc_modem	2	2	ISO 15118 PLC modem	2 candidate	Resolve whether these allocations are one shared physical item or separate install locations before sourcing, costing or deduplicating.
mid_energy_meter	2	2	MID energy meter	2 candidate	Resolve whether these allocations are one shared physical item or separate install locations before sourcing, costing or deduplicating.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
AC input breaker and SPD	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
lockable AC isolator	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
surge arrester cartridge	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
input terminal shroud	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
DC output contactor set	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
precharge resistor module	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
DC output fuse carrier	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
output polarity label	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
MID energy meter	critical unpriced	2 allocations	critical	unpriced					Resolve duplicate component identity before sourcing: decide whether this is one shared physical item or separate allocated lines.
meter CT shunt set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
meter seal kit	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
meter calibration label	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Power module stack	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
power module backplane	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
module locking rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
DC-link capacitor bank	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
DC output choke	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
EMI filter assembly	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
snubber network board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
thermal interface pad set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
module health monitor board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
heatsink temperature probe	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
fan/coolant derating table	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
firmware recovery jumper	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
CCS2 liquid-cooled cable and connector	critical unpriced	unique	critical	unpriced					Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
connector temperature sensor pair	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
proximity pilot resistor set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cable strain relief boot	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
liquid-cooled cable manifold	candidate unpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
quick-connect coolant coupler	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
coolant leak catch tray	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cable coolant temperature probe	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
ISO 15118 PLC modem	critical unpriced	2 allocations	critical	unpriced					Resolve duplicate component identity before sourcing: decide whether this is one shared physical item or separate allocated lines.
control pilot interface board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
proximity detection input	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
vehicle comms isolation transformer	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
charge controller PCB	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
safety IO terminal block	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
real-time clock module	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
nonvolatile event log memory	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.

Showing first 40 of 120 ledger rows; export CSV for the full line-by-line ledger.

BoM Provenance Manifest

0 source-backed claims, 52 missing-source claims, 0 provenance violations. Critical missing-source claims: 52. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
AC input breaker and SPD	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
AC input breaker and SPD	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
AC input breaker and SPD	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
AC input breaker and SPD	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC output contactor set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC output contactor set	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC output contactor set	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC output contactor set	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Power module stack	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Power module stack	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Power module stack	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Power module stack	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CCS2 liquid-cooled cable and connector	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CCS2 liquid-cooled cable and connector	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CCS2 liquid-cooled cable and connector	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
1	CCS2 liquid-cooled cable and connector	missing source			evidence before BoM can pass.
					Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
2	ISO 15118 PLC modem	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
3	ISO 15118 PLC modem	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
4	ISO 15118 PLC modem	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
5	ISO 15118 PLC modem	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
6	OCCP communications gateway	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
7	OCCP communications gateway	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
8	OCCP communications gateway	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
9	OCCP communications gateway	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
ISO 15118 PLC modem	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
ISO 15118 PLC modem	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
ISO 15118 PLC modem	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
ISO 15118 PLC modem	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MID energy meter	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Insulation monitoring device	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Insulation monitoring device	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Insulation monitoring device	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Insulation monitoring device	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Cooling loop assembly	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Cooling loop assembly	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Cooling loop assembly	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Cooling loop assembly	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Emergency stop and safety interlock set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Emergency stop and safety interlock set	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Emergency stop and safety interlock	mpn	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
set					supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Emergency stop and safety interlock set	unit_cost_gbp	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Insulation monitoring device	supplier	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Insulation monitoring device	manufacturer	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/120 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 7 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:power_distribution_ac_input_switchgear_ac_input_breaker_and_spd.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
AC input breaker and SPD	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
lockable AC isolator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
surge arrestor cartridge	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
input terminal shroud	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
DC output contactor set	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
precharge resistor module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									nt planning starts.
DC output fuse carrier	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
output polarity label	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
MID energy meter	duplicate identity review required	critical	unpriced			0/4	no	no	Resolve duplicate component identity before pricing or using this line.
meter CT shunt set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
meter seal kit	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
meter calibration label	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
Power module stack	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
power module backplane	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
module locking rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
DC-link capacitor bank	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
DC output choke	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
EMI filter assembly	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
snubber network board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
thermal interface pad set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
module health monitor board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
heatsink temperature probe	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
fan/coolant derating table	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
firmware recovery jumper	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
CCS2 liquid-cooled cable and connector	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
connector temperature sensor pair	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
proximity pilot resistor set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cable strain relief boot	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
liquid-cooled cable manifold	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
quick-connect coolant coupler	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
coolant leak catch tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cable coolant temperature probe	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
ISO 15118 PLC modem	duplicate identity review required	critical	unpriced			0/4	no	no	Resolve duplicate component identity before pricing or using this line.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
control pilot interface board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
proximity detection input	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
vehicle comms isolation transformer	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
charge controller PCB	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
safety IO terminal block	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
real-time clock module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									sourcing is complete or procurement planning starts.
nonvolatile event log memory	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
OCPP communications gateway	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
LTE router module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
ethernet surge protector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
secure element module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
ISO 15118 PLC modem	duplicate identity review required	critical	unpriced			0/4	no	no	Resolve duplicate component identity before pricing or using this line.
Plug and Charge certificate store	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
charging session firmware image	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
protocol debug header	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
MID energy meter	duplicate identity review required	critical	unpriced			0/4	no	no	Resolve duplicate component identity before pricing or using this line.
DC voltage transducer	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurem ent	Required Action
									procurement planning starts.

Showing first 50 of 120 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

120 closure row(s): 7 ready, 6 blocked, 107 deferred. Actions: 7 collect source, 0 repair, 0 replace protocol, 107 candidate deferrals. Procurement blockers 13; priced-review blockers 13; next row bom-close:charging_connector_interface_ccs2_cable_connector_ccs2_liquid_cooled_cable_and_connector.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	CCS2 liquid-cooled cable and connector	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	OCPP communications gateway	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	Power module stack	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	Cooling loop assembly	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	AC input breaker and SPD	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	DC output contactor set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	Emergency stop and safety interlock set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	ISO 15118 PLC modem	blocked	blocker	resolve component identity	duplicate identity review required	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated componentWorldId or quantity basis	yes	yes	Resolve duplicate component identity before pricing or using this line.
9	ISO 15118 PLC modem	blocked	blocker	resolve component identity	duplicate identity review required	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated componentWorldId or quantity basis	yes	yes	Resolve duplicate component identity before pricing or using this line.
10	MID energy meter	blocked	blocker	resolve component identity	duplicate identity review required	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed	yes	yes	Resolve duplicate component identity before pricing or using this line.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
						lines; updated componentWorldd or quantity basis			
11	Insulation monitoring device	blocked	blocker	resolve component identity	duplicate identity review required	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated componentWorldd or quantity basis	yes	yes	Resolve duplicate component identity before pricing or using this line.
12	MID energy meter	blocked	blocker	resolve component identity	duplicate identity review required	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated componentWorldd or quantity basis	yes	yes	Resolve duplicate component identity before pricing or using this line.
13	Insulation monitoring device	blocked	blocker	resolve component identity	duplicate identity review required	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated componentWorldd or	yes	yes	Resolve duplicate component identity before pricing or using this line.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
						quantity basis			
14	cable strain relief boot	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	connector temperature sensor pair	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	proximity pilot resistor set	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	cable coolant temperature probe	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
18	coolant leak catch tray	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
19	liquid-cooled cable manifold	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	quick-connect coolant coupler	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	control pilot interface board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	proximity detection input	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
23	vehicle comms isolation transformer	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
24	ethernet surge protector	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	LTE router module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	secure element module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	charge controller PCB	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
28	nonvolatile event log memory	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
29	real-time clock module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	safety IO terminal block	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	charging session firmware image	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	Plug and Charge certificate store	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
33	protocol debug header	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
34	fan/coolant derating table	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	firmware recovery jumper	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	heatsink temperature probe	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	module health monitor board	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
38	DC output choke	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
39	EMI filter assembly	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	snubber network board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	thermal interface pad set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	DC-link capacitor bank	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
43	module locking rail	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
44	power module backplane	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	coolant pump	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	coolant reservoir	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	plate radiator	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
48	airflow proving switch	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
49	cabinet fan tray	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	exhaust louvre assembly	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 120 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

3 batch(es): 1 active, 1 waiting, 1 deferred. Active rows 7; deferred rows 107. Critical source rows 7; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:ccs2_liquid_cooled_cable_and_connector.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
1	Resolve Component Identity Before Pricing	waiting	blocker	ISO 15118 PLC modem	ISO 15118 PLC modem; ISO 15118 PLC modem manufacturer part number; ISO 15118 PLC modem catalogue price GBP	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated component WorldId or quantity basis	yes	component identity unambiguous
1	Resolve Component Identity Before Pricing	waiting	blocker	ISO 15118 PLC modem	ISO 15118 PLC modem; ISO 15118 PLC modem manufacturer part number; ISO 15118 PLC modem catalogue price GBP	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated component WorldId or quantity basis	yes	component identity unambiguous
1	Resolve Component Identity Before Pricing	waiting	blocker	MID energy meter	MID energy meter; MID energy meter manufacturer part number; MID energy meter catalogue price GBP	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated component WorldId or quantity basis	yes	component identity unambiguous
1	Resolve Component Identity	waiting	blocker	Insulation monitoring device	Insulation monitoring device;	canonical component identity	yes	component identity

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
	Before Pricing				Insulation monitoring device manufacturer part number; Insulation monitoring device catalogue price GBP	decision; whether repeated allocations are one shared physical item or separate installed lines; updated component WorldId or quantity basis		unambiguous
1	Resolve Component Identity Before Pricing	waiting	blocker	MID energy meter	MID energy meter; MID energy meter manufacturer part number; MID energy meter catalogue price GBP	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated component WorldId or quantity basis	yes	component identity unambiguous
1	Resolve Component Identity Before Pricing	waiting	blocker	Insulation monitoring device	Insulation monitoring device; Insulation monitoring device manufacturer part number; Insulation monitoring device catalogue price GBP	canonical component identity decision; whether repeated allocations are one shared physical item or separate installed lines; updated component WorldId or quantity basis	yes	component identity unambiguous

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	CCS2 liquid-cooled cable and connector	CCS2 liquid-cooled cable and connector; CCS2 liquid-cooled cable and connector manufacturer part number; CCS2 liquid-cooled cable and connector catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	OCPP communications gateway	OCPP communications gateway; OCPP communications gateway manufacturer part number; OCPP communications gateway catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Power module stack	Power module stack; Power module stack manufacturer part number; Power module stack catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Cooling loop assembly	Cooling loop assembly; Cooling loop assembly manufacturer part number;	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					Cooling loop assembly catalogue price GBP			
2	Collect Critical Source Evidence	active	blocker	AC input breaker and SPD	AC input breaker and SPD; AC input breaker and SPD manufacturer part number; AC input breaker and SPD catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	DC output contactor set	DC output contactor set; DC output contactor set manufacturer part number; DC output contactor set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Emergency stop and safety interlock set	Emergency stop and safety interlock set; Emergency stop and safety interlock set manufacturer part number; Emergency stop and safety interlock set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	cable strain relief boot	cable strain relief boot; cable strain relief boot manufacturer part number; cable strain relief boot catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	connector temperature sensor pair	connector temperature sensor pair; connector temperature sensor pair manufacturer part number; connector temperature sensor pair catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	proximity pilot resistor set	proximity pilot resistor set; proximity pilot resistor set manufacturer part number; proximity pilot resistor set catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	cable coolant temperature probe	cable coolant temperature probe; cable coolant temperature probe manufacturer part number; cable coolant temperature probe catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-	deferred	minor	coolant leak catch tray	coolant leak catch tray; coolant leak	no immediate evidence	no	candidate remains out of

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
	Only Sourcing				catch tray manufacturer part number; coolant leak catch tray catalogue price GBP	required unless this candidate is promoted to procurement scope		procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	liquid-cooled cable manifold	liquid-cooled cable manifold; liquid-cooled cable manifold manufacturer part number; liquid-cooled cable manifold catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	quick-connect coolant coupler	quick-connect coolant coupler; quick-connect coolant coupler manufacturer part number; quick-connect coolant coupler catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	control pilot interface board	control pilot interface board; control pilot interface board manufacturer part number; control pilot interface board catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	proximity detection input	proximity detection input; proximity detection input manufacturer part number; proximity detection input catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	vehicle comms isolation transformer	vehicle comms isolation transformer; vehicle comms isolation transformer manufacturer part number; vehicle comms isolation transformer catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	ethernet surge protector	ethernet surge protector; ethernet surge protector manufacturer part number; ethernet surge protector catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	LTE router module	LTE router module; LTE router module manufacturer part number; LTE router module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/13; procurement-blocking closure rows 13; active sourcing rows 7; deferred sourcing rows 107. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_review_required. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.17	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/13 critical rows production-eligible; 7 critical unsourced, 0 protocol-only, 0 source-reference blocked.	power_distribution_ac_input_switchgear_ac_input_breaker_and_spd: Collect admissible source evidence for this critical line before BoM can pass. power_distribution_dc_output_switchgear_dc_output_contactor_set: Collect admissible source evidence for this critical line before BoM can pass. power_distribution_billing_metering_branch_mid_energy_meter: Resolve duplicate component identity before pricing or using this line. energy_conversion_transduction_rectifier_power_modules_power_module_stack: Collect admissible source evidence for this critical line before BoM can pass. charging_connector_interface_ccs2_cable_connector_ccs2_liquid_cooled_cable_and_connector: Collect admissible source evidence for this critical line before BoM can pass. plus 8 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.8917	13 procurement-blocking closure row(s); 7 active sourcing row(s), 107 deferred row(s).	bom-close:charging_connector_interface_ccs2_cable_connector_ccs2_liquid_cooled_cable_and_connector: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_compute_communication	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				_backend_gateway_ocpp_communications_gateway: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_conversion_transduction_rectifier_power_modules_power_module_stack: Collect admissible source evidence for this critical line before BoM can pass. bom-close:environmental_interface_liquid_cooling_loop_cooling_loop_assembly: Collect admissible source evidence for this critical line before BoM can pass. bom-close:power_distribution_ac_input_switchgear_ac_input_breaker_and_spd: Collect admissible source evidence for this critical line before BoM can pass. plus 8 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.		Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.		Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0	costing_not_started ; critical priced 0/13; BoM total 0 GBP.		Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				power_distribution_billing_metering_branch_mid_energy_meter: Resolve duplicate component identity before sourcing: decide whether this is one shared physical item or separate allocated lines. energy_conversion_transduction_rectifier_power_modules_power_module_stack: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. charging_connector_interface_ccs2_cable_connector_ccs2_liquid_cooled_cable_and_connector: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 79 more	

prototype procurement policy	review	0	candidate_bom_review_required; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.
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Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
AC input breaker and SPD	1 candidate	AC input breaker and SPD; AC input breaker and SPD manufacturer part number; AC input breaker and SPD catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
DC output contactor set	1 candidate	DC output contactor set; DC output contactor set manufacturer part number; DC output contactor set catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
MID energy meter	1 candidate	MID energy meter; MID energy meter manufacturer part number; MID energy meter catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Power module stack	1 candidate	Power module stack; Power module stack manufacturer part number; Power module stack catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
CCS2 liquid-cooled cable and connector	1 candidate	CCS2 liquid-cooled cable and connector; CCS2 liquid-cooled cable and connector manufacturer part number; CCS2 liquid-cooled cable and connector catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
ISO 15118 PLC modem	1 candidate	ISO 15118 PLC modem; ISO 15118 PLC modem manufacturer part number; ISO 15118 PLC modem catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
OCPD communications gateway	1 candidate	OCPD communications gateway; OCPD communications gateway manufacturer part number; OCPD communications gateway catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
ISO 15118 PLC modem	1 candidate	ISO 15118 PLC modem; ISO 15118 PLC modem manufacturer part number; ISO 15118 PLC modem catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
MID energy meter	1 candidate	MID energy meter; MID energy meter manufacturer part number; MID energy meter catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Insulation monitoring device	1 candidate	Insulation monitoring device; Insulation monitoring device manufacturer part number; Insulation monitoring device catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Cooling loop assembly	1 candidate	Cooling loop assembly; Cooling loop assembly manufacturer part number; Cooling loop assembly catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Emergency stop and safety interlock set	1 candidate	Emergency stop and safety interlock set; Emergency stop and safety interlock set manufacturer part number; Emergency stop and safety interlock set catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Insulation monitoring device	1 candidate	Insulation monitoring device; Insulation monitoring device manufacturer part number; Insulation monitoring device catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
IEC-61851	Conductive charging system requirements	UK/EU	Charging modes, control pilot and safety-function evidence.
IEC-62196-3	DC vehicle connector requirements	UK/EU	CCS2 connector rating, cable assembly and temperature evidence.
ISO-15118	Vehicle-to-grid communication interface	Global / EU	PLC communication, Plug and Charge and session-state evidence where applicable.
OCPP-1.6J/2.0.1	Charge point backend protocol	Operator requirement	Backend session, metering, fault and firmware-management evidence.

Hazard	S	O	D	RPN	Mitigation
DC output fault or insulation failure	10	3	3	90	Insulation monitoring, contactor isolation and fault-state logging.
Connector overheating at high current	8	4	4	128	Temperature sensing, derating and liquid-cooled cable verification.
Backend or payment outage stops sessions	5	4	4	80	Offline policy, local fault handling and backend retry logic.

Section Issues And Score

bom

- blocker / critical_part_unpriced: AC input breaker and SPD is critical but has no credible unit cost.
- blocker / critical_part_unpriced: DC output contactor set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: MID energy meter is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Power module stack is critical but has no credible unit cost.
- blocker / critical_part_unpriced: CCS2 liquid-cooled cable and connector is critical but has no credible unit cost.
- blocker / critical_part_unpriced: ISO 15118 PLC modem is critical but has no credible unit cost.
- blocker / critical_part_unpriced: OCPP communications gateway is critical but has no credible unit cost.
- blocker / critical_part_unpriced: ISO 15118 PLC modem is critical but has no credible unit cost.
- blocker / critical_part_unpriced: MID energy meter is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Insulation monitoring device is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Cooling loop assembly is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Emergency stop and safety interlock set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Insulation monitoring device is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

[illegible]

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
Mean	7.67						